



A Robotic Mind Model for Affective Decision Making and Behaviour Generation

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Abstract

Affective robots are of interest for many applications, to accompany people, assist in education, and reassure patients. However, the principles for control of such a robot are not yet established, and its specifications are not clear. Here we propose a framework for a lively robot that can reason to make decisions and maintain emotions and intentions. The Affective RObotic Mind (AROM) model is designed and implemented, aiming to enhance the interpretability of affective robot behaviour for humans in various social situations encountered by the robot. Inspired by biological examples, AROM integrates robot decision-making and behaviour-generation modules based on consistent affective support for taking action based on decisions and expressing internal states via body movements. The core factors that adjust decision-making and movement patterns are abstracted from the physiological system that is shared by most vertebrates. The system is thus intended as an exemplary architecture for biomimetic affective robot design. To evaluate the achieved affective behaviour in the robot, experiments on human subjects are conducted related to the function of the decision-making module and the emotion-related enhancement of interpretability of the behaviour-generation module. The project exemplifies the potential of biomimetic design of robot emotion expressions and its promising value in shaping robots' personality and behaviour logic.

Keywords Human-robot interaction · Social robots · Affective robots · Decision making · Behaviour interpretability

1 Introduction

Questions such as: What kind of robots do we humans want? Is it necessary to assign emotions to robots to improve human acceptance of robots? What other benefits can affective robots bring to humans? are still critical in the field of human-robot, and various approaches have been proposed to address these issues. Stock et al. [1, 2] believes that affective robotics can reduce human cognitive burdens in Human-Robot Collaboration workspace. Cabibihan et al.

[3] assert that social robots enable significant benefits in the therapy of autistic children, due to predictability and adaptability. Based on such evidence, the related research field Affective Computing also benefits, which introduces new approaches also into general Robotics by the development of a dedicated field of *Affective Robotics*.

Despite multiple attempts were made in the past decades, the theories and methods of Affective Robotics remain still underexplored. For an emotion model functioning within a robot system, there is no agreed approach connecting decision-making and behaviour generation. Furthermore, a general control logic for emotional robot control is unavailable. Nevertheless, studies have applied basic discrete emotion models to robotics systems [4–6] and achieved meaningful results, although restricted to the generation and control of movement, because of the nature defects of discrete models on control issues. Research on generating behaviours is mostly focused on producing facial expressions [6, 7], and the studies that do enable robots to use other body parts for emotional expression are usually lacking adaptability and just mimicking human emotional behaviour or that of

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a specific animal species [5]. Aiming to tackle these problems, this project proposed AROM, an Affective Robotic Mind model for emotion-intervening decision making and behaviour generation. It provides a general, adaptive and consistent solution for emotion model, robot behaviour-generating and decision making, with enough interpretability for humans to understand.

The AROM system applies a biomimetic model using continuous emotion that generalises the widely used PAD (Pleasure, Arousal, Dominance) emotion model [8]. The emotion system is inspired by the emotion theory proposed by Rolls [9], which establishes a continuous relationship between emotions and various types of reinforcement contingencies. One reason for referring to this is its better applicability to Reinforcement Learning as compared, e.g., to Pekrun's Control-Value Theory [10] and Ekman's Basic Emotions Theory [11]. The emotion model of AROM is thus able to fluctuate emotions according to the rewards and context information, enabling being utilised in decision making of learning tasks. In the behaviour-generating module, AROM defines several affective motion parameters, as well as an algorithm for generating behaviours with only robot body parameters required, and can be applied in various types of robots. In addition, the decision-making module is able to conduct computations in an emotionless as well as in emotional mode where emotions may be subject to ongoing change. This module is formulated as an MDP, for details see below. All parts of the system share the same affective framework, and are able to combine and generate complex motions under affective decisions and posture expressions. In order to evaluate the performances of the AROM system, human experiments are conducted to examine perceptions of emotions that the robot reveals. Simulations of robot decision making when facing an unexpected adversity are also implemented under different settings of the emotion model. The aim is to test whether emotions can help robots adapt to the changing environment.

Focusing on the AROM system, two questions will be explored in this study:

1. Does emotion have benefits in addition to the facilitation of communication and of agreement among agents? We will consider here in particular the adaptation and interaction of a robot to its environment.
2. Can a map from robotic emotions to robotic behaviour be established that shares general features with most of the vertebrates? Although there is currently no agreement on the design principles for robotic emotions, such a map can help to reduce the arbitrariness in the design of robotic emotions.

The overall contribution of this project is to propose an integrative framework for affective robot decision-making and behaviour generation, the results of which are adaptable and interpretable. Related works of this project are demonstrated and discussed in Chapter 2. In Chapter 3, the methods of this project are illustrated, with Table 1 showing the list of symbols used in this paper. Chapter 3.2 describes the emotion model and Chapter 3.3 explains the decision-making module. In Chapter 3.4, the behaviour-generation module is discussed. Chapter 4 reports on the experiments, with decision-making in Chapter 4.1 and behaviour-generation in Chapter 4.2. Further aspects of this project are discussed in Chapter 5, and conclusions of this study are drawn in Chapter 6.

2 Related Work

2.1 Background

With the general increase in robotic applications, also pet robots, educational and care robots will be in higher demand, which creates the necessity to develop methods that provide the robots in these domains with affective intelligence, to perceive human emotions and make sensible decisions as well as demonstrate emotional behaviours. Affective robots are designed not to rely purely on logic to decide about the actions, for example, pet robots should be programmed to accommodate flexibly a variety of goals under identical external conditions in order to provide entertaining and engaging companionship [12]. This can include even mildly provocative and irrational behaviour, as people may prefer a robotic pet that may not be optimal, predictable or passive-reactive, but that engages in emotional communication.

To enable robots to interact with humans in complex environments, cognitive models have been developed for robots to improve their overall performance and to achieve acceptance by the human partner. Cognitive models that have been tested successfully in robots include ACT-R [13], SOAR [14], and CRAM-A [15], while other models with similar capabilities, such as LIDA have not received the same attention by the robotics community. These cognitive models simulate different aspects of cognitive modules such as perception, navigation, action and memory. However, flexible adaptability, intrinsic motivation that we would expect in a robotic pet, is often not discussed in existing cognitive models. This led us to consider emotional dynamics in cognition systems.

The AROM system focuses on the cognitive functions related to emotions, especially the effect of emotions and the assessment of various parameters of the emotional model. While emotions were also studied (e.g., in SOAR

Table 1 List of symbols in the methodology section

Symbols	Meaning
Parameters for emotional dynamics and interactions	
t	Time
E_t, E_0	Level of affect dimension of at time t , level of original affect
P	Level of PLEASURE
A	Level of AROUSAL (Noradrenaline)
D	Level of DOMINANCE (Serotonin)
DA	Level of DOPAMINE
R	Stimulus value
S^A	Level of satiation
k_e, k_{da}	Damping constant for E, DA
k_p, k_a, k_d	Damping constants for P, A, D
m_p, m_a, m_d	Sensitivity of P, A, D to changes of R
κ	Sensitivity of DA to changes of the value of satiation S^A
h	Gain of satiation from the rewards
q	Number of time steps of historical information used in the calculation of D
Parameters for decision making	
W	Weight value for general states in the MDP
W_{idle}	Weight value for idle state specifically
W_{bs}	Original basic weight value without influences of affect
$W_{bs-idle}$	Original basic weight value for idle state
λ	Degree that DOPAMINE DA can influence the values of weight W
μ	Degree of Serotonin (DOMINANCE) D influencing weights W
δ	Level of Serotonin (DOMINANCE) D influencing weights for IDLE state W_{idle}
U	Threshold for Noradrenaline (AROUSAL) A
$\alpha_{x_t}, \beta_{x_t}$	Parameters of the beta function in Thompson Sampling (TS), see Algorithm 2
r_t	Reward in Thompson Sampling (TS), see Algorithm 2
P	Transition probability in the MDP
P_{TS}	Basic probability generated by Thompson Sampling (TS)
Parameters for behaviour generation	
H_{Sp}	Effect of PLEASURE P on suppression
H_{Sd}	Effect of DOMINANCE D on suppression
H_{Va}	Effect of AROUSAL A on velocity
H_{Vd}	Effect of DOMINANCE D on velocity
H_{Ap}	Effect of PLEASURE P on amplitude
H_{Aa}	Effect of AROUSAL A on amplitude
H_{Ad}	Effect of DOMINANCE D on amplitude
H_{Fa}	Effect of AROUSAL A on frequency
v_{Sd}	Damping constant of DOMINANCE D regarding to suppression
v_{Vd}	Damping constant of DOMINANCE D regarding to velocity
v_{Ad}	Damping constant of DOMINANCE D regarding to amplitude
v_{Va}	Damping constant of AROUSAL A regarding to velocity
v_{Aa}	Damping constant of AROUSAL A regarding to amplitude
v_{Fa}	Damping constant of AROUSAL A regarding to frequency
K	Requirement of triggering Primary Locomotion Level behaviours, see Algorithm 3

[16, 17]), emphasising the influence of emotions in specific cognitive architectures, an analysis of the functions and features of emotions on different cognitive modules was not

central to their argument. Taking here a different perspective enables us also to leverage benefits for applications in Human-Robot Interaction.

2.2 Robotic Emotion Models

Most of the Affective Robotics studies utilise emotion models from psychology. Commonly used models can be divided into two types: discrete models and dimensional models which are also known as continuous models. The most known discrete emotion model is Paul Ekman's theory of six basic patterns of expression (happiness, anger, disgust, fear, sadness, and surprise) [11]. With emotional parameters being clearly defined and categorised, Ekman's model is applied in many robotics studies and helps the generation and recognition of emotional behaviours. However, as a discrete model, the expression space of it is limited compared with dimensional models, where emotions can be described continuously using dimensions.

The existing dimensional model can have distinctive numbers of dimensions, but the most widely applied one in robotics is the PAD emotional state model developed by Mehrabian [8]. It uses three numerical dimensions of Pleasure, Arousal and Dominance to express all the emotional states. They measure the degree of pleasant, energized, and controlling feelings, respectively. Another three-dimensional model is Lövheim cube [18], which is an explanatory model for emotions and monoaminergic neurotransmitters. Lövheim picked monoamines Dopamine, Serotonin and Noradrenaline as the three dimensions and illustrated the emotions arisen with fluctuations of the three monoamines. In [18], the three monoaminergic dimensions are measured based on eight basic emotions. For instance, under low levels of Dopamine and Serotonin, the rise of the Noradrenaline level matches the change from emotion shame to distress. Whereas, these two emotional models both have disadvantages on Human-Robot Interaction practice. It is hard to find traces and research on how emotions influence decision-making process and provide robotics a good reference if only based on the PAD model. Under the construction of Lövheim cube and monoamine, though there is sufficient evidence on how monoaminergic neurotransmitters influence decision-making process, the empirical study of the relationship between Lövheim's emotion model and behaviour-generation is very rare. We resolve this problem by abstracting featured functions of monoamines and combining them with emotional dimensions that are massively used in behaviour experiments, and enable our model to support both decision-making and behaviour generation for robots.

2.3 Affective Decision Making

Historically, most research on affective robot decision-making was carried out by coalescing physiological features to decision-making systems. Lewis et al. [19] proposed a robot

architecture to investigate the roles of Pleasure in the action selection process in survive tasks. During the decision-making tasks, the pleasure hormone alters the incentive salience of perceived stimuli by acting on the 'subjective assessment' or 'assignment of value'. Cos et al. [20] designed a homeostatic adaptive mechanism based on RL to modulate the internal deficits of a social agent. The robot tries to maintain a good physiological state by learning the optimised actions in variant situations, and the physiological dimensions of Tiredness and Hunger evolve over time. Together these studies provide important insights into the functions of physiological factors in survive tasks, whereas the affect dimensions involved are limited, unable to describe more complex states of robots.

A more comprehensive set of basic emotions is used in [21]. It includes hope, fear, happiness and sadness. In this study, an affective model for decision agents in non-expensive robotic platforms is demonstrated. Under the Adversarial Risk Analysis framework, the agent makes decisions according to emotional factors in a human-robot interaction scenario. However, this affective model is mainly oriented towards decision-making in robots, and the emotional dimensions are insufficient to be used in comprehensive behaviour generation, which requires more affective factors and cannot function well based on vague states of hope, specifically. Motivated by establishing a framework which both possesses physiological features of affective decision-making, and also utilises the same emotion model which supports robot behaviour-generating, we designed the decision-making module of AROM, inspired by functions of three monoamines and supports updating emotions for generating behaviours.

2.4 Affective Behaviour Generation

Many research works on affective robot behaviour generation are conducted by mimicking human behaviours. Suguitan et al. [22] proposed a method for automatically synthesizing affective robot movements based on generative adversarial neural networks (GANs) [23]. This method utilises human behaviours to generate robot motions that match the affective features of human movements. Ali et al. [5] focused on affective non-verbal human-robot interaction, applying 28 overlapping emotions including 8 primary categories based on Russell's circumplex model [24], which is a circumplex model sorting 28 emotions using the two bipolar dimensions of valence and activation. However, such studies focus only on robot body part postures and movements, without paying much attention to whole body trajectory generation (such as the trajectory of the robot's walking) and the affective control under different levels of the same emotions.

To fill this gap, [25] studied robot motion and kinematics variables' impact on people in the practice of navigation. The movements are specified by velocity profiles using periodic perturbation as variants to the profiles. Results of it show that using saccadic or smooth motion can impact the perception of the emotions of the robot, such as confidence, frailty, and aggression. Nevertheless, the trajectory movements in this study simply support moving in a straight line. To enable richer affective navigation expression as well as suitable body movements, we include a behaviour-generating module into the proposed AROM, which also aims at promoting adaptivity to more general types of robots, although humanoid robots are not considered at this stage.

2.5 The AROM Model and Comparisons to Other Models

In this paper, we tried to fill the gap in the research of robotic emotion model, decision-making and behaviour generation. The AROM model combines PAD emotion model and Lövheim cube of monoamine, providing a continuous model for emotions and building a link between emotions and external stimuli for emotional fluctuation, which is applicable to both decision-making and behaviour generating process. The decision-making module of AROM can not only deal with the incentive salience of stimuli and conduct "subjective assessment" on the value or reward of tasks, it can also regulate the robot's belief in the form of probability, making it easily combined with popularly used methods in robotics such as Reinforcement Learning. The usage of three-dimensional emotion model (AROUSAL (noradrenaline), DOMINANCE (serotonin) and DOPAMINE) in decision-making process also make it possible for the robot to react to wider range of emotions, not restricted to specific basic emotions like hope, fear, happiness and sadness. In behaviour generation module of AROM, we also make progress in the range of emotions that can be referred to, and the types of behaviours generated. It surpasses the restriction of the research before which generates behaviours based on classic human behaviour features, supporting continuous emotion controls, and can be used to generate not only body part behaviours but also whole body trajectory moves. In addition, our model supports generating behaviours under different levels of the same emotions, and relative experiments have been conducted to evaluate to what degree humans can perceive robots' emotion intensity.

Although more experiments still need to be done to explore the boundaries of our model as well as its functions and influences in applications, AROM model provides a novel solution for the whole system design for Affective Robotics and tackles the problems that previous methods

designed for single affective modules are incompatible. Compared with other models, our model's contributions are:

1. Propose a novel solution to the whole system design for Affective Robotics, instead of only targeting to one aspects of robotics like decision-making or behaviour generating.
2. Our four-dimensional continuous emotion model supports a wider range of emotional control, and a link between emotions and external stimuli for emotional fluctuation has been built.
3. Through "subjective assessment" on values and belief probabilities in decision-making, our model can fit in many decision-making methods with these two factors such as Reinforcement Learning.
4. Being able to generate both body part behaviours and whole body moving trajectory, our model supports generate behaviours with different emotion dimensions as well as intensity, and also have potentials to be applied on distinctive robot types.

3 Methodology

The AROM (Affective RObotic Mind) model is intended to integrate decision-making and behaviour-generation under affective modulation and to enable a robot to work with genuine emotions. It is inspired by the monoamine neurotransmitter system shared by virtually all vertebrates, functioning to promote the adaptability to various environments [26, 27]. Monoamines like dopamine, serotonin and noradrenaline adjust physiological reactions, and influence the motivations and emotions on the levels of both decision making and behaviour adjustment. Targeting similar functions, the AROM model provides an affective robot control architecture, enabling the module of decision making and behaviour generation be integrated, and influenced together by specific emotional states at the current time.

Different from traditional methods in robotics, which usually compute decisions of actions and design emotional motion patterns separately, the AROM model supports display of consistent affects on both of these levels, and enables robots to react to environmental stimuli in a vivid and quasi-natural way. The hypothesis is, under AROM model, affective robots can be perceived as more emotionally prominent and their intentions are easier to understand.

As illustrated in Fig. 1, the AROM model is composed of the following three main components.

- *Emotion Calculation*: Core module of the AROM model, it influences both the decision making and Behaviour Generation modules.

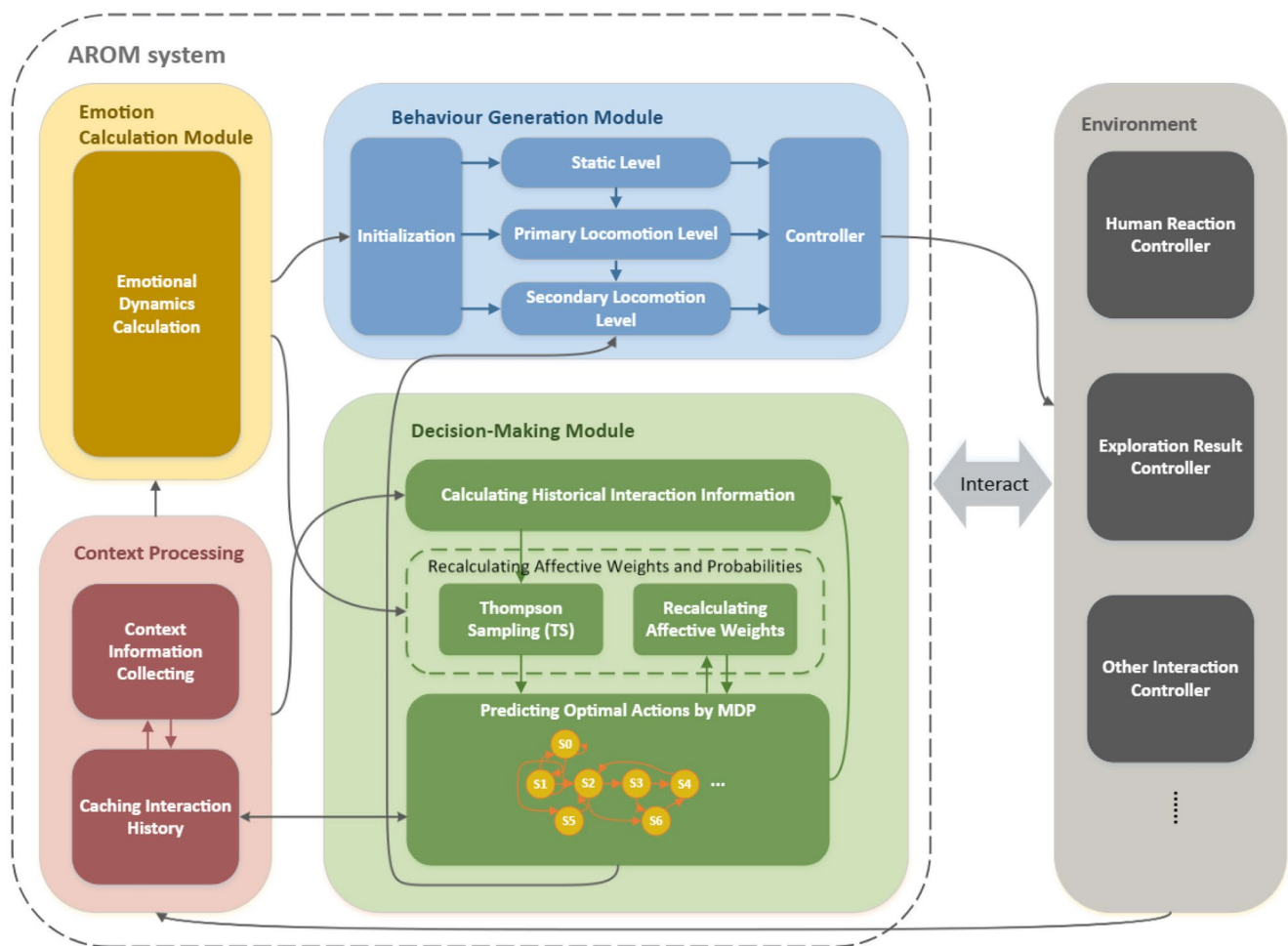


Fig. 1 The global system structure of the Affective Robotic Mind (AROM) model is proposed here, featuring modules for emotion calculation, behaviour generation, and decision making. Interaction with

the environment is based on sensory input entering via context processing unit and behavioural output via the behaviour generation module. For details, see the respective sections below

- **Decision Making:** Mainly based on a Markov Decision Process (MDP), it utilises parameters of physiology and emotion to decide structural actions such as approaching the human partner or playing alone.
- **Behaviour Generation:** Controlled by physiological and emotional parameters, it generates behaviour details based on the decisions according to certain motion parameters like velocity, force and stretching level.

In addition to its main components, the model makes use of a *Context Processing* module and an *Environment* module, which are also shown in Fig. 1. The *Context Processing* and *Environment* module are not major contents of this paper, and are separated to help readers understand how the information of context and environment is processed in AROM system.

- **Context Processing:** Continuously collects context information and caches interaction history, which directly

influence emotion calculation and engages in the decision-making process.

- **Environment:** Shows the origin of possible interaction results, such as human reaction controller and exploration result controller. The system keeps interacting with the *Environment* in the human-robot interaction scenario.

3.1 Functions of Emotions in AROM

Inspired by the three-dimensional monoamine and emotion model [18], the emotion model of AROM merges the monoamine model and PAD emotion model. The monoamines like Dopamine, Serotonin and Noradrenaline engage the adjustment and regulation of robot decision-making and behaviours, just as they do in the vertebrate body.

The PAD emotion model is frequently used in affective robotics, because it is convenient to refer to enabling an easier comparison of different standards of emotional effects.

Although just as [18] declared, the three-dimensional monoamine model does not fully match the PAD dimensional emotion model, and there exists rotation of model comparing of two of these. Furthermore, the pleasantness dimension is commonly taken in many emotion models, while the monoamine model [18] does not utilise it, we tried to fill this gap in this paper.

In the AROM model, we construct an emotion model possessing features from both the monoamine system and the PAD emotion model. In order to simplify the model and to reduce the number of dimensions, the SEROTONIN dimension in the monoamine model is merged to the DOMINANCE dimension in the PAD model, and the Noradrenaline dimension is merged to the AROUSAL.

As illustrated in Fig. 2, the original monoamine dimensions are used to adjust the process of decision-making in the robot, while the original PAD emotion dimensions are to regulate parameters in the behaviour generation process. From the perspective of the monoamine model, the DOPAMINE dimension mainly influences the motivation level when making decisions, and the Serotonin (DOMINANCE) dimension adjusts the confidence and satisfaction of the decision-making process, while the Noradrenaline (AROUSAL) level has effects on making rational or risky choices. From the point of view of the PAD emotion model, the PLEASURE dimension can influence behavioural patterns as well as some motion parameters, such as speed. The AROUSAL dimension adjusts the degree of excitement of a movement by regulating parameters like speed, acceleration, time gap to change motion patterns, and force. Finally, the DOMINANCE (Serotonin) level influences the posture of behaviour generation, e.g., shrinking body when losing DOMINANCE.

The emotions applied in the decision-making module are DOPAMINE, Serotonin (DOMINANCE) and Noradrenaline (AROUSAL). Their functions are abstracted from the realistic functions of those monoamines on decision making. The monoamine dopamine plays an integral role in coding rewards in decision making [28], and during decision making, it controls the incentive salience function, which triggers a pulse of “wanting” towards the cues of rewards

[29, 30]. We incorporate the incentive salience as the main feature of DOPAMINE in the AROM system and utilise it to adjust the weights in the MDP.

Serotonin functions to constrain the response of external stimuli and adjusts ongoing processing of sensory input [31]. Considering that the mechanism and influences of it are very complex, we only borrow parts of features of it overlapping with emotional dimension dominance, i.e. facilitating motor output and suppressing ongoing processing of sensory input during motor behaviour [31, 32], and apply DOMINANCE to adjust action weights. In a more intuitive way, it can be understood as that DOMINANCE controls the robot acting actively or in depression.

Noradrenaline is believed to reduce the usage of decision-relevant information when choosing actions. Evidence also suggests that noradrenaline results in more random decisions instead of a directed exploration [33, 34]. We will therefore include in our model a similar negligence to relevant information and a tendency to make more random decisions, enabling an impact of AROUSAL on the probabilities in the MDP.

Both physiological observations and empirical relationships between emotions and movements are considered in the design of the behaviour-generating processes in AROM. Starting with physiological behaviours, they are influenced by monoamines, especially by serotonin, which affects muscular tension and causes the “5-HT posture” when injected into animals. This has been described as “holding a stance elevated above the substrate with the abdomen flexed under its body” [35, 36]. Inspired by this, we design the DOMINANCE to influence the STRETCH and SHRINK of the robot’s body. Aside from this, we abstract emotional behaviour features from affective gait research [37–39]. Dynamic emotional parameters that participate in calculation are PLEASURE, AROUSAL (Noradrenaline) and DOMINANCE (Serotonin). The general functions of them towards generating robot behaviours are illustrated in Table 2. As the value of PLEASURE rises, it will impose an impact on robot behaviours to reduce the inhibition of movement, and lift the angle of swinging arms or other similar types

Fig. 2 Monoamine-based emotion-behaviour model. (a) original PAD model [8] including monoamine-based interpretation. (b) l  vheim cube of the relationship between the monoamine neurotransmitters and emotions [18]. In both models the information relevant for the AROM model (see Fig. 1) is added

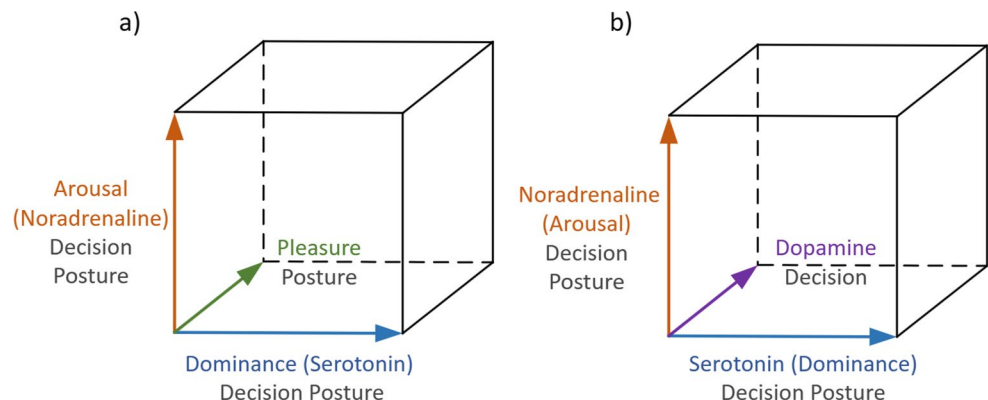


Table 2 General influences of emotions to motions. Influences (+) refers to means the effects of emotions occurring at positive values, and influence (-) to negative values. The arrows $\uparrow$ and $\downarrow$ indicate an increasing or decreasing effect on behaviour dimensions, respectively

	Influences (+)	Influences (-)
PLEASURE	Behaviour inhibition $\downarrow$	Behaviour inhibition $\uparrow$
	Swing angle $\uparrow$	Swing angle $\downarrow$
AROUSAL	Gait velocity $\uparrow$	Gait velocity $\downarrow$
	Swing $\uparrow$	Swing $\downarrow$
	Thigh angle $\uparrow$	Thigh angle $\downarrow$
	Step length $\uparrow$	Step length $\downarrow$
DOMINANCE	Default	Inhibiting reactions

of body. As PLEASURE value reduces, it will have a reversed influence. With the AROUSAL value increases, for situations where robots have legs, the gait velocity, swing movement, thigh angle, and step length will increase, and vice versa. For robots that do not have legs and move otherwise, the impact on aspects of legs will be altered correspondingly. At a higher value of DOMINANCE, we assume that the current robot movement is retained by default. However, at a lower value, DOMINANCE has the effect of inhibiting reactions of the robot both regarding speed and amplitude.

3.2 Emotion Model and Response to Stimuli

Because emotions are not always static, the calculation of how emotions fluctuate and respond to stimuli is significant, in particular for a robot that can react to the external world and has various emotional expressions. During the update of emotions, two components are to be considered: one is the natural decay of emotions, and the other relates to the change of emotions when external stimuli arrive and interfere with the affective system.

For the damping of emotions, we use an exponential function to mimic the kinetics of monoamine elimination:

$$E_t = E_0 e^{-k_e t} \quad (1)$$

Here E denotes the level of one of the affect dimensions, which can be DOPAMINE, Serotonin (DOMINANCE), Noradrenaline (AROUSAL) or PLEASURE. E_0 is the base affect

level, and k_e is the rate constant of damping. For a list of all symbols used here, see Table 1.

To tailor to the MDP-based affective decision-making process (see Sect. 3.3), the variation of emotions can be expressed as:

$$E_t = E_{t-1} e^{-k_e} \quad (2)$$

The time t in eq. (1) is initialised by a value of 1, which also equals the time step in the MDP dynamics.

For changes of emotions when exposed to various external stimuli (see Table 3), the four dimensions are under the influence of the positive or negative nature of stimuli, the change rate of stimuli, and the fluctuation of satiation. Succeeding the reinforcer-emotion association hypothesis that Rolls [9] proposed, the emotion model in AROM system introduces reinforcers and stimuli to help regulate the variations of PAD emotions. The symbol $S+$ refers to a positive stimulus, and $S-$ means a negative one. The symbol stands for termination, e.g., $S+!$ means the termination of a positive stimulus. $S+ - \downarrow$ means that the stimulus intensity decreases, regardless of whether the stimulus is positive or negative. Similarly, $S+ - \uparrow$ implies an increase of the stimulus intensity. In this paper, we use rewards R as in reinforcement learning [40] to evaluate stimuli S . For instance, a reward with negative value can be taken as a negative stimulus $S-$ and vice versa.

The levels of emotion can be determined accordingly. For example, PLEASURE is expected to decrease, if the robot is exposed to negative stimuli and to increase, if the robot is exposed to positive stimuli. The dynamics of PLEASURE P can thus be modelled by

$$P_t = P_{t-1} e^{-k_p} + m_p (R_t - R_{t-1}) \quad (3)$$

where P is the degree of pleasantness, $e^{-k_p t}$ is a damping factor, m_p governs the sensitivity to stimuli, and R is the value of stimuli, which is here referred to as *reward*, analogous to the reward concept in reinforcement learning [40]. The level of PLEASURE only depends on the temporal changes of stimuli and on the intrinsic damping, so we only compare the reward of the current and of the previous time.

Table 3 The response to stimuli and satiation state in the emotion dynamics. The symbol S represents stimuli. The symbol $S+$ refers to a positive stimulus, and $S-$ to a negative one. The symbol stands for termination, e.g., $S+!$ means the termination of a positive stimulus. $S+ - \downarrow$ means that the stimulus intensity decreases, regardless of whether the stimulus is positive or negative. Similarly, $S+ - \uparrow$ implies an increase of the stimulus intensity. Sat refers to satiation as a physiological state. Since the three-dimensional monoamine system and PAD emotion model are merged here, some of the names of monoamines are shown in brackets. 5-HT represents serotonin, and NA represents Noradrenaline

Various Situations and Affects								
Stimulus/Satiation	S-	S+	S+ - $\downarrow$	S+ - $\uparrow$	S+!	S-!	Sat+	Sat-
Direction of Change	-	+	-	+	-	+	-	+
Emotional States		PLEASURE		AROUSAL (NA)		DOMINANCE (5-HT)		DOPAMINE

The degree of AROUSAL (Noradrenaline) is influenced by the intensity of any sudden change on the stimulus. If the changes are always slow, then the level of AROUSAL will gradually decay, and if the changes are large and quick, it will rise. Here, the level of AROUSAL A is determined by the following relation,

$$A_t = A_{t-1}e^{-k_a} + m_a|R_t - R_{t-1}| \quad (4)$$

where m_a governs the sensitivity of the AROUSAL dimension of the stimuli. The damping acts in the same way in the dynamics of PLEASURE in equation 3. Because the changes of AROUSAL are generally sudden, we only consider the absolute changing value of rewards between the current and the previous time step.

The dynamics of DOMINANCE is a simplified model of the serotonin system and is affected by the gap between expectation and reality. Here, we use the termination of positive or negative stimuli to govern DOMINANCE by comparing the changing rate of historical stimuli and new stimuli. For instance, if the robot always receives positive stimuli, but suddenly a negative stimulus arrives, then the DOMINANCE decreases by a small amount, expressing in a sense of expectation that the situation is starting to deteriorate. Similarly, if at one time the robot receives a stimulus that is more positive than usual, the robot may feel more confident about this task. The update for DOMINANCE D is thus given by

$$D_t = D_{t-1}e^{-k_d} + m_d \left(\frac{d(R_t - R_{t-1})}{dt} - \frac{d(R_{t-1} - R_{t-q})}{dt} \right) \quad (5)$$

where m_d controls the sensitivity to stimuli, q determines the range of historical information used in the update, and the damping is similar to the previous affect definition. Because the change of DOMINANCE is related to the difference of the new stimulus from the old ones, in Eq. 5 the changes of reward between the current and last time and several previous times are compared.

The dynamics of DOPAMINE is in a system distinct from the aforementioned affect dimensions, and since the AROM model mainly utilises the function of DOPAMINE to regulate motivation, here DOPAMINE is influenced by the level of various forms of satiation. If the robot has low satiation with respect to the battery level or social contact, then DOPAMINE will increase and provide higher motivation for the robot to reach a higher level of satiation and to obtain what it wants. If the robot is more satisfied with the present conditions, then the DOPAMINE level will decrease and the robot will be less inclined to achieve those goals. The calculation function for DOPAMINE DA is:

$$DA_t = DA_{t-1}e^{-k_{da}} + \kappa S_t^A \quad (6)$$

where DA is the level of DOPAMINE, with S^A representing the level of satiation. Inspired by research on incentive salience of DOPAMINE in the field of computational biology [29], the rate parameter κ represents the sensitivity of satiation in the present context.

The satiation S^A is set related to the stimuli of specific tasks. For task x , we represent the related stimulus value as R_t^x at time t , which can be modelled as rewards. The calculation function for the satiation level of task x is:

$$S_t^A = \begin{cases} hR_t^x, & \text{if } R_t^x > 0 \\ 0, & \text{if } R_t^x < 0 \\ -1, & \text{if } R_t^x = 0 \end{cases} \quad (7)$$

where h is the parameter controlling the gain of satiation from the rewards. In this model, when negative outcomes happen, the satiation level keeps no change. When no actions are conducted related to the task (the gained reward is 0), the value of satiation is set to -1 enabling the robot to accumulate “wanting” for the task after a period of time not conducting it. If the robot tries on the task but fails (i.e., receives a negative reward), we keep the satiation unchanged.

3.3 The Decision-Making Model

The decision-making process of the AROM model is influenced by both the interaction context information and the affect factors. As illustrated in Fig. 1, the decision-making module is composed of two procedures, one of which is calculating weight values and probabilities, and the other one employs an MDP-like dynamics to determine decision probabilities. The computations in this module are based on the context and on historical information, and specifically the weight values are calculated based on emotional and cognitive WANTING factors. After decision making, the decided action will impact the behaviour generation module, along with the calculated affect value.

For a biomimetic decision-making module, the goal is to enable the robot to decide reasonably and in a way similar to a human (or another animal). As a simple realisation of such a module, we refer to Markov Decision Process (MDP), which were employed as a basic model for human decision-making process in neuroscience [41] and psychology [42]. The key components of an MDP are the states, actions, transition probabilities and rewards. The agent interacts with the environment by taking actions to transit from state to state with certain transition probabilities, and obtain according rewards. Here, the MDP model is embedded in the AROM

model which interferes with the Markov property and the matrix-linearity of the state transitions. Because of its basic structure and its relevance in the theoretical study of decision making, we will still refer to it as an MDP even if this holds in a strict sense only for the isolated system.

We proceed using the original concept of “reward” as a weight value, which represents how the robot values the states to reach, as a combination of rational judgement, its physiological “wanting”, and affective influence to the states. The probability distribution, which is the probability of taking action for given current states to next states as originally declared in Reinforcement Learning, represents the “understanding” of the world by the robot, e.g., to what probability can the robot reach this state from the last state by taking this action?

AROM supports general types of task-specific MDP, and emotional states or actions do not need to be included since rewards and transfer probabilities are all that emotions adjust. However, in the experiments of this paper, the states of the MDP model in use are modified to represent the positive or negative features of the consequences of actions in order to illustrate the relationship between states and

resulted emotions more clearly. For instance, if the robot takes action to approach a human, the corresponding state depends on the result, namely whether it is accepted by the human and in an ensuing interaction with the human as anticipated, or whether it is refused. By defining states with these features, the effect of stimuli can be easily determined, as shown in Section 3.2, for instance, after some actions the robot reaches the state with a negative feature, then it can be assumed as receiving one kind of negative stimuli. The weight value here for each state can be seen as the degree that this state is positive or negative. As a result, the weight value can not only be utilised to engage in the decision-making process through the MDP model, but also can be utilised to calculate the updated affects, as declared in Section 3.2.

3.3.1 The Decision-Making Process

The process of MDP decision-making sub-module is shown in Fig. 3. Starting from calculating historical interaction information, the module recalculates affective weights and probabilities. After that, the affective probability weights are

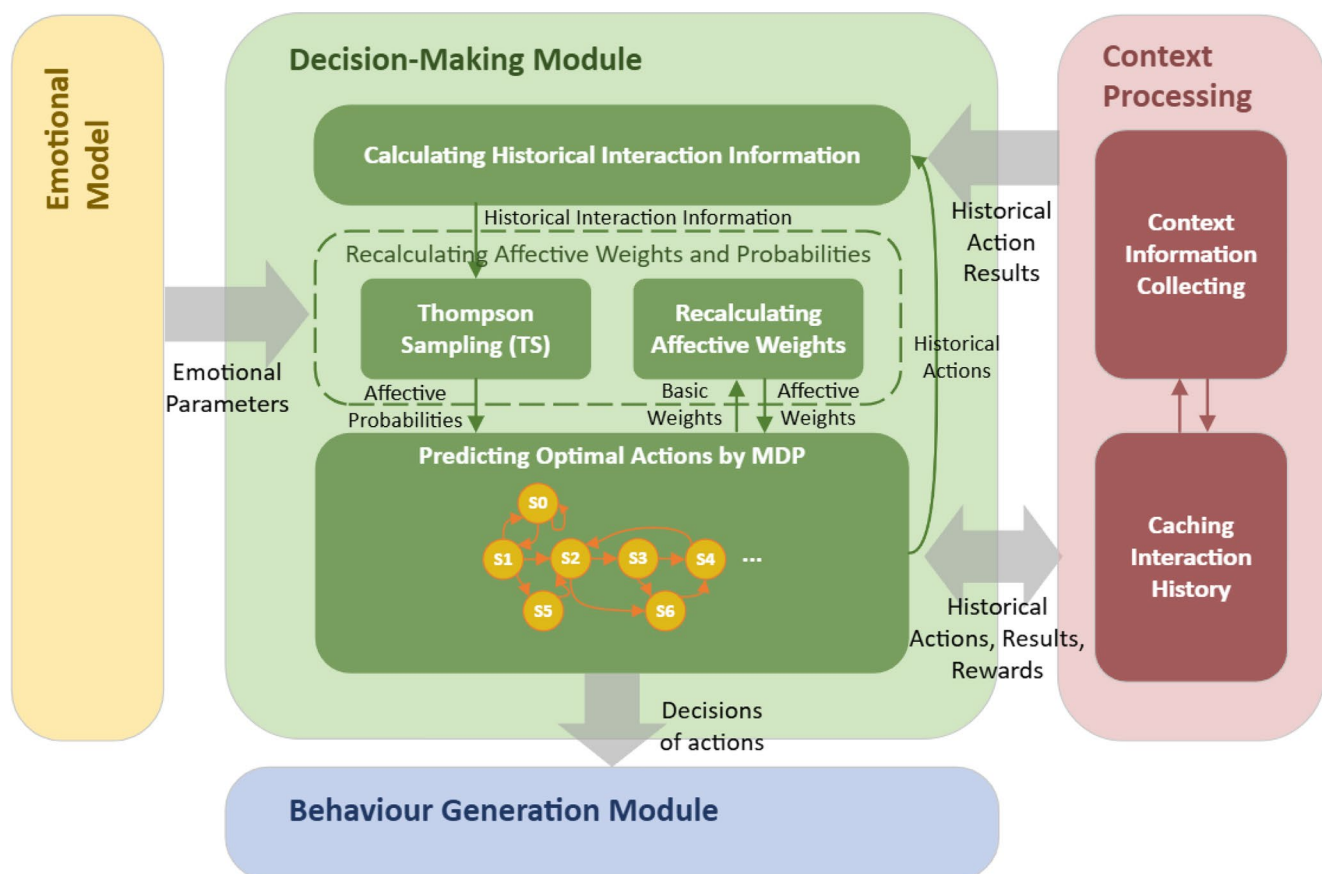


Fig. 3 The decision-making module is based on an MDP and supports general types of task-specific MDP. It does not necessarily include emotional states or actions. This module receives inputs from the emo-

tion module and is affected by context information. In this way, the weights (or rewards in the case using MDP) and transition probabilities are dynamic under adjustments of emotions on longer time scales

transferred to MDP, and utilised to predict optimal actions with emotions. The MDP uses Value Iteration Algorithm 1 to calculate the best action to take.

After the decision is made, the robot will conduct this decision and interact with the external environment, and receive the results of this interaction. The results will be matched to the states with positive or negative features, and then the robot will be assumed at this matched state, and use this state as the “last state” for the next MDP best action calculation. During this process, each time the robot obtains interaction results from outside, it will calculate them as historical information and cache them. From this information, the robot calculates the historical transition probability distribution, which is the expectation of the robot whether specific actions can be successful or are likely to fail. This distribution will be introduced into the MDP model as the updated transition probability distribution for the next MDP calculation.

Algorithm 1 Value Iteration Algorithm [40, 43]. s represents states, a actions, s' represents the subsequent state reached from s , r is the reward, P is transition probability, $Q(s, a)$ is the expected cumulative reward when applying action a in states, and $V(s)$ denotes the value of state s .

```

Initialise  $V(s)$  by arbitrary values
Repeat until  $V(s)$  converges
for all states do
  for all actions do
     $Q(s, a) \leftarrow \sum_{s'} P_{ss'}^a [r(s, a, s') + \gamma V(s')]$ 
  end for
   $V(s) \leftarrow \max_a Q(s, a)$ 
end for

```

The calculation of historical transition probability distribution uses Thompson Sampling (TS) to avoid frustration caused by cold boot and significantly bad responses in a constantly changing system. TS is used to model the uncertainty or belief about the reward probability of each arm in multi-armed bandit problems, and is also used as a heuristic solution of the exploitation-exploration dilemma [44, 45]. The beta distribution provides a way to represent this uncertainty in a probabilistic manner with two parameters α and β . The more rewards obtained relative to selections, the higher the shape parameter α will be, reflecting a higher estimated reward probability. Here, TS is not directly applied and controls the robot to choose the action with the highest estimated reward probability. It calculates the *reward probability* as the transition probability in the MDP functioning as the robot’s understanding of the world. As the robot has a higher historical probability to obtain a certain state S , the belief of getting S at the next action becomes higher and more stable. But if the robot is continuously refused by the

human, for example, and cannot reach a certain state S by taking action a , then TS will affect a lower belief and thus prompt exploration of other actions. The details of TS are shown in Alg. 2.

Algorithm 2 Thompson Sampling in AROM (K, α, β) [46]. t represents time, k represents the number of actions, $\hat{\theta}_k$ is the estimated success probability of action k , $beta(\alpha_k, \beta_k)$ is the beta distribution with two parameters: α and β , x represents the action and P is the distribution of probability, r represents the reward.

```

for  $t=1, 2, \dots$  do
  for  $k=1, \dots, K$  do
    Sample  $\hat{\theta}_k \sim beta(\alpha_k, \beta_k)$ 
  end for
   $P_{x_{kt}} \leftarrow \max_k \hat{\theta}_k$ 
   $r_t \leftarrow$  MDP state transition using  $P_{x_{kt}}$ 
   $(\alpha_{x_t}, \beta_{x_t}) \leftarrow (\alpha_{x_t} + r_t, \beta_{x_t} + 1 - r_t)$ 
end for

```

3.3.2 Calculation of Intervening Affects

Based on the aforementioned MDP decision-making system, the affective factors are calculated and combined during the decision-making process. There are two main types of variables that influence how decisions are produced, which are the weight values and transition probabilities. The weight values represent which states the robot considers as important, whether it “wants” them and to what extent it needs them; the transition probability distribution is the robot representation of the world, describing whether it is possible to reach a certain state S by conducting action A . Here, we show how the emotions function on these two variables, and thus influence the decision-making process.

The weight value of the MDP is mainly affected by the factors of DOPAMINE and Serotonin (DOMINANCE). One of the functions of DOPAMINE is controlling the “wanting” parameter, and the larger the “wanting” parameter, the higher the weight value is to be set in making a decision. Another factor that influences decision-making is Serotonin (DOMINANCE). Just consider how Serotonin is used to treat depression: through increasing the intensity of Serotonin, patients can become more energetic. We can analogously assign a higher a weight value to the states (with the exception of IDLE), if with a larger Serotonin factor, representing that robots are activated to make decisions and move around. Since the serotonin functions on suppression of sensory input and constraining behavioural activation are more obvious and classical, we only design the serotonin influence when it is at negative values in this paper.

In this paper, we calculate two types of weight values. One is the weight value for general states W , and the other

one is the weight value for idle states W_{idle} . The IDLE state is a special state where the robot keeps idle, and usually happens when the robot has a low level of Serotonin (DOMINANCE). It can be understood as the robot's tendency to keep idle due to the influence of serotonin to constrain activity and cause depression.

The calculation of weight values for general states is illustrated below:

$$W = \begin{cases} W_{bs} e^{\lambda DA} e^{\mu D}, & \text{if } D < 0 \\ W_{bs} e^{\lambda DA}, & \text{if } D \geq 0 \end{cases} \quad (8)$$

where W represents the weight value for general states, W_{bs} represents the original basic weight value without influences of affects. DA is the value of DOPAMINE, and λ represents the degree that DOPAMINE can influence the weight values; D is the value of Serotonin (DOMINANCE), and similarly μ represents the degree of it to influence weights. The effects from Serotonin (DOMINANCE) D are only valid when D is negative, and if D is neutral or positive the weight keeps as default. For weights of IDLE particularly, the calculation is:

$$W_{\text{idle}} = \begin{cases} \frac{W_{bs-\text{idle}}}{\delta e^D}, & \text{if } D < 0 \\ W_{bs-\text{idle}}, & \text{if } D \geq 0 \end{cases} \quad (9)$$

where W_{idle} represents weight value for the IDLE state specifically, and $W_{bs-\text{idle}}$ represents the original basic weight value for the IDLE state. D is the value of Serotonin (DOMINANCE), and δ is the level that Serotonin (DOMINANCE) can influence weights for the IDLE state. It should be emphasised that IDLE is not identical to sleeping or resting, and that we temporarily ignore the “wanting” of the robot to rest.

The transition probability distribution is mainly under the impact of Noradrenaline (AROUSAL). Noradrenaline, as far as researched, can influence rational and irrational decisions, and one reason that it causes this change is that it may slow the reaction to new changes of the environment, e.g., not being sensitive to new information and making decisions according to old learned knowledge and natural “wanting”. Thus, we utilise this feature of Noradrenaline (AROUSAL) to modify the transition probability distribution in the MDP model, by adjusting the parameters of α and β in Thompson Sampling (TS) (Alg. 2), as well as normalising the probability to reduce the importance of the success probability and raise targets that are more *wanted*. While Noradrenaline (AROUSAL) A exceeds a positive threshold value U , the calculation of transition probability is:

$$(\alpha_{x_t}, \beta_{x_t}) \leftarrow \left(\alpha_{x_t} + \frac{r_t}{A+1}, \beta_{x_t} + 1 - \frac{r_t}{A+1} \right) \quad (10)$$

$$P = P_{\text{TS}} - A \left(P_{\text{TS}} - \frac{1}{2} \right) \quad (11)$$

where $\alpha_{x_t}, \beta_{x_t}$ are the parameters from TS, r_t represents rewards in TS, A represents the value of Noradrenaline (AROUSAL), P represents the probability, P_{TS} is the basic probability, which is generated through TS, which can be influenced by affects or not based on the value of affect. Equation 10 illustrates that in TS, with Noradrenaline (AROUSAL) A increasing, the rewards r_t added to beta function in TS will be decreased in updating, so as to make the robot less sensitive to past successes or failures. Equation 11 shows the normalisation of P from a range of $[0, 1]$ towards $\frac{1}{2}$ (here $A > U$, and $U > 0$, $A \in [0, 1]$), representing relevant ignorance of success probability. The higher A is, the closer the probability approaches 0.5; and if $A = 0$, then the probability keeps not change as the basic probability P_{TS} .

3.4 The Behaviour-Generating Process

In addition to allowing for emotional intervention in decision-making, affective robots require a module to generate interpretable and vivid behaviours. The importance of a proper design of robots' emotional expression in behaviours has been studied and confirmed by several researchers. Tsiourti et al. [47], for example, argue that any incongruence of robotic emotional expression will confuse human observers and may result in the robot being perceived as unintelligent or unsympathetic. As proposed by Koschate et al. [48], the displays of emotions can reduce the uncanniness of human-like robots.

Beyond helping humans to develop positive impressions of robots, we aim in this project at establishing a general and adaptive behaviour-generating mechanism for robots. This mechanism is designed based on continuous emotional states, and can be adopted for various types of robots and for a variety of tasks. Through adjusting motion parameters according to emotional dynamics, the behaviour generating process of AROM provides an architecture for mapping and generation of various emotions and robot behaviours, including large-scale trajectory generation, control of body posture and possibly also of detailed distal movements. The generating model is inspired by physiological observations and the empirical relationship between emotions and movements, and it aims to be compatible with human intuition and to reach a level of generality that enables comparisons between machines and humans.

3.4.1 Affect-Intervened Motion Parameters

Table 2 shows the simplified relationship model between emotional dynamics and robot behaviours. Based on this

model, the affect-intervened motion parameters are calculated on two dimensions: Static Posture and Body Movement. For computational convenience, the emotional influences on Static Posture are combined as a new dimension: Deviation, as shown in Table 4. The motion parameters related to the dimension of Static Posture are STRETCH and SHRINK, which describe the direction of robot static postures, and STRETCH represents a direction outwards from the body, while SHRINK represents direction towards the inside.

For instance, for the head and neck part of postures, STRETCH means the robot raising its neck and chin up, while SHRINK refers to a downward movement of the head onto the chest (as if the robot intends to shrink its body). For the ears, for example, STRETCH means spreading the ears outwards, and SHRINK means turning the ears inwards. The parameters of STRETCH and SHRINK compose the basic variables for controlling robot static postures, and they are combined to a united control parameter: SUPPRESSION. The parameter SUPPRESSION describes the static postures of movable joints analogous to the definition for STRETCH and SHRINK. In terms of robot movements, SUPPRESSION describes the medium (or basic) degree of movement. The eyelids of the robot, for example, may be static under some conditions, but can partially close under certain emotions to indicate a level of tiredness. However, if the robot is required to blink eyes, the eyelids will be dynamic and move upwards and downwards, so that the medium degree of movement appears as the static posture of eyelids, if they do not need to move, and this specific posture is defined by SUPPRESSION. Usually, according to the changing emotions the robot is going through, the SUPPRESSION parameter is not directly related to the mechanically neutral position of a posture. For instance, the lower value of PLEASURE may cause an increase of SUPPRESSION, which causes a deviation of the median positions from the neutral positions. Under the BODY MOVEMENT dimension there are three motion parameters, which are VELOCITY, AMPLITUDE and FREQUENCY respectively. Just as explained by their names, they described the velocity, amplitude and frequency of robot

body part movements as well as whole body moving trajectory by walking, running or crawling. For example, for swinging movements of the arm of the robot, the velocity of swing is controlled by VELOCITY, and the scale of swing is controlled by AMPLITUDE. If the swing movement continuously exists or happens regularly, then the frequency of the movement is controlled by FREQUENCY. Similarly for whole body moving trajectory, VELOCITY controls the velocity of walking, AMPLITUDE for moving range and FREQUENCY may influence the frequency the robot changes its direction when walk aimlessly. Expanded from Table 2, here in Table 4 the relationship between emotions and the motion parameters is illustrated. The emotion of PLEASURE influences the STRETCH, SHRINK, SUPPRESSION and AMPLITUDE. For positive values of PLEASURE (+), the STRETCH value increases, resulting the SUPPRESSION value decreasing, and increasing the AMPLITUDE value. For negative values of PLEASURE (−), the SHRINK value decreases, resulting in an increase of the SUPPRESSION value, while the AMPLITUDE value decreases. The AROUSAL value influences the VELOCITY, AMPLITUDE and FREQUENCY parameters. For positive values of AROUSAL (+), the VELOCITY, AMPLITUDE and FREQUENCY increase, while for negative values of AROUSAL (−), the three parameters decrease. The level of DOMINANCE affects the parameters of STRETCH, SUPPRESSION, VELOCITY and AMPLITUDE, but the influences are only effective when the DOMINANCE value is negative (−). The negative DOMINANCE values will cause the SHRINK to decrease, thus leading to an increase of SUPPRESSION, with VELOCITY and AMPLITUDE decrease.

Based on Table 4, motion parameters are calculated according to emotional parameters of PLEASURE, AROUSAL and DOMINANCE. The effects of those emotional parameters are assumed that: PLEASURE has linear effects on behaviour parameters; AROUSAL has stronger influences at larger values, and the effects are nonlinear; only when DOMINANCE is in its negative range it has influences on behaviours, and the influences are in a form of nonlinear damping. Under such kind of hypothesis, the motion parameters are calculated for SUPPRESSION, VELOCITY, AMPLITUDE, and FREQUENCY:

Table 4 Framework of generating emotional robot behaviours. Emotion of Pleasure, Arousal, and Dominance influences both static posture and body movement of robots. For computational convenience, the influences of emotions on static postures are programmed to a new dimension called Deviation, which can also be applied when generating moving behaviours

Behaviour		Static Posture		Deviation	Body Movement		
		STRETCH	SHRINK	SUPPRESSION	VELOCITY	AMPLITUDE	FREQUENCY
PLEASURE	+	↑		↓		↑	
	−		↓	↑		↓	
AROUSAL	+				↑	↑	↑
	−				↓	↓	↓
DOMINANCE	+						
	−		↓	↑	↓	↓	

$$S = \begin{cases} H_{Sp} \left(\frac{P+1}{2} \right) \times H_{Sd} e^{v_{sd}D}, & \text{if } D < 0 \\ H_{Sp} \left(\frac{P+1}{2} \right), & \text{if } D \geq 0 \end{cases} \quad (12)$$

$$V = \begin{cases} H_{Va} e^{v_{Va}(A-1)} \times H_{Vd} e^{v_{Vd}D}, & \text{if } D < 0 \\ H_{Va} e^{v_{Va}(A-1)}, & \text{if } D \geq 0 \end{cases} \quad (13)$$

$$A = \begin{cases} H_{Ap} \left(\frac{P+1}{2} \right) \times H_{Aa} e^{v_{Aa}(A-1)} \times H_{Ad} e^{v_{Ad}D}, & \text{if } D < 0 \\ H_{Ap} \left(\frac{P+1}{2} \right) \times H_{Aa} e^{v_{Aa}(A-1)}, & \text{if } D \geq 0 \end{cases} \quad (14)$$

$$F = H_{Fa} e^{v_{Fa}(A-1)}, \text{ for all } D \quad (15)$$

where in Equation 12, H_{Sp} is the degree at which PLEASURE P influences suppression, with H_{Sd} representing the degree of influence from DOMINANCE D . v_{sd} is the damping constant of DOMINANCE D regarding to suppression.

Equations 13, 14, and 15 are understood in the same way, where the first uppercase subscript characters of H and v represent the motion parameters, V for velocity, A for amplitude and F for frequency. The second lowercase subscript characters describe the emotional dimensions the effects from, with p for PLEASURE, a for AROUSAL and d for DOMINANCE. The reason for applying $\frac{P+1}{2}$ and e^{A-1} is to regularise the ranges of calculated motion parameters, which are restricted to $[0, 1]$, considering that the range of the quantities P , A , D is $[-1, 1]$.

3.4.2 Behaviour Generating Algorithm

The behaviour-generating algorithm of AROM utilises the motion parameters calculated from emotions to let the robot produce affective behaviours, which can be combined with and controlled by the decision-making module, being able to adapt to distinctive types of behaviours. The inputs of the algorithm are parameters of robot systems like types of joints, scales of movement, and neutral positions of each joint or movable unit. After being calculated with the real-time motion parameters which contain emotional information, they transfer to the control parameters of robots, and finally generate affective robot behaviours.

As shown in Fig. 4, the behaviour-generating module starts from an initialisation phase, and then calculates robot behaviour across several levels, from the Static Level, the Primary Locomotion Level to the Secondary Locomotion Level, and finally sends the target behaviour parameters to the controller of the robot. At the initialisation stage, several inputs including emotional parameters and the basic parameters of the robot movement are received and processed, with movement parameters being affected also by emotion parameters.

In more detail, the three levels of behaviour generation are as follows:

- *Static Level*: to generate static postures, which change all the time with emotions

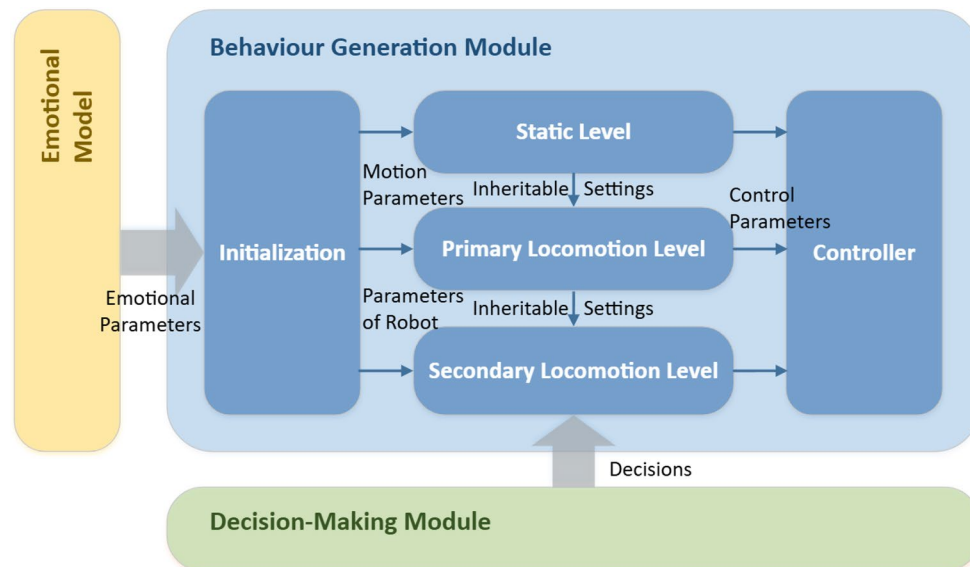


Fig. 4 Structure of the behaviour-generating process. The module starts from initialization of emotional parameters and basic robot motion parameters, it then calculates robot behaviour across static level, Primary Locomotion level and Secondary Locomotion level, and finally sends control parameters of generated behaviour to the controller. The three levels of behaviour generation are: 1) Static level: to generate

static postures with emotions. 2) Primary Locomotion level: to generate emotional movements without engagement of decisions from decision-making module. 3) Secondary Locomotion level: to generate important movements that are mainly required by decision-making module with emotions

- *Primary Locomotion Level*: Generating movement parts that are not very significant for robots to complete tasks or so, but still have meanings such as expressing emotions. The robot components at this level will start to move or stop depending on triggers of specific strong emotions etc., and can be understood as almost fully controlled by the behaviour-generating module.
- *Secondary Locomotion Level*: Generating important movements that are mainly required by the decision-making module, of which some movement parameters may change according to emotions, but the joint or body parts on this level will keep moving or stop under control based on decision making.

While generating behaviours, these three levels of behaviour generation inherit settings from their previous level, and make use of movement features that were calculated already. For instance, for a movement belonging to the Primary Locomotion level, a requirement may exist for the robot to remain static for a certain interval of time, in which case the calculated Static-Level behaviours can be directly applied and sent with the movement generation output of the primary level to the controller. The analogous effect can be seen at the Secondary Locomotion Level, where the behaviour generation process can refer to the results of the decision-making module. In intervals between two decisions, where such results are unavailable, the behaviours of the primary level can be applied.

Based on this structure for behaviour generating, the algorithm is given as pseudo-code in Algorithm 3. As outlined in the algorithm, the inputs contain emotional parameters, neutral positions and movable scales of the joints of the robot, neutral velocity and velocity scale calculated from the max velocity of the robot, a basic continuous movement pattern function such as the trigonometric function for eyelid opening, tail wagging and so on, and the decisions made by the decision-making module. Through calculation, the algorithm outputs the parameters and functions for behaviour control, both for the movement trajectory of the robot and for movements of body appendages. On the Static Level, the joint positions are adjusted according to motion parameter SUPPRESSION. When a requirement for Primary Locomotion Level is met, which is here usually the presence of certain emotions, the robot starts to conduct locomotion. The motion parameter FREQUENCY adjusts how frequently the robot conducts the locomotion, and the body movement amplitude is modified by the AMPLITUDE parameter, with VELOCITY modifying the linear velocity and AMPLITUDE any angular velocities. It should be noted that here the velocities do not directly relate to the velocities of the movements of body parts. Instead, they are defined at the level of full-robot trajectory, although for complex robots sometimes the

velocity needs to be inferred from the velocities of robot body parts, according to the way the robot moves. For the Secondary Locomotion Level, the behaviour generation is based on the framework of the primary level, combined with decisions, which may assign the target movement directions for robots, depending on what decision was taken.

The algorithm is adaptive to variable task types. The movement parameters and patterns for each joint can be designed specifically, and at the same time still show affective features during the movements of the robot. For special robot body parts that not suit the framework of this algorithm, the appliers also can pick appropriate motion parameters of SUPPRESSION, VELOCITY, AMPLITUDE and FREQUENCY and design their own unique affective behaviours.

4 Experiments

In order to assess the AROM model and the functions of emotions in decision-making and behaviour generation, experiments in human-robot interaction scenarios are conducted. For decision-making, we are curious about how emotions can influence the robot's learning of the changes of environment, and deal with sudden and large fluctuations of emotions. Thus, the experiment scenario where the robot needs to learn and deal with the human's sudden change of attitudes toward playing (the human first always accepts playing, and then always refuses) is chosen. For behaviour generation, we want to know to what degree humans can perceive robots' emotions from their physical motions. Behaviours with variant emotions are generated based on the rabbit-like robot MiRo and recorded, then human subjects are required to watch and mark their perceptions and feelings towards these motions.

4.1 Decision Making and Emotion Modules During Interaction

4.1.1 Experimental Design

As many physiologists proposed, physiology is about how the living organism adjusts to the adversities of the environment, and enable them to adapt to the changing world [49, 50]. The organism escapes from dying cold or heat, moves to find suitable surroundings, water and food. They are quickly aroused if meeting predators, and calm down at their safe spaces. Inspired by such features, the decision-making module is designed. Thus, in the experiment, we test the adaptivity of the module to the changing environment, and the benefits of emotions in robot decision-making in the scenarios of human-robot interaction.

Algorithm 3 Behaviour-Generating Algorithm

Input: emotional parameters P , A , D , neutral positions and movable scales N_j , S_j for j th movable joints of the robot, neutral linear and angular velocity N_l , N_a and their scales S_l , S_a , basic continuous movement pattern function $T_j(t)$ at time t , decision H

Output: static affective behaviour parameters B_j and movement function $B_j(t)$ for j th movable joints of the robot, linear and angular velocity v_l , v_a

Initialise motion parameters Sup , Vel , Amp , $Freq$ from P , A , D , requirement K for Primary Locomotion Level

for all j **do**

$B_j \leftarrow N_j - 2(Sup - 0.5)S_j$

▷ Static Level

end for

if K is met **then**

for t per $\frac{1}{Freq}$ **do**

for all j **do**

$B_j(t) \leftarrow T_j(t) \times Amp$

▷ Primary locomotion level

end for

$v_l \leftarrow N_l + 2(Vel - 0.5)S_l$

$v_a \leftarrow N_a + 2(Amp - 0.5)S_a$

if H exists **then**

for all j **do**

Update $B_j(t)$ based on $T_j(t) \times Amp$ and H

▷ Secondary locomotion level

end for

Update v_l based on $N_l + 2(Vel - 0.5)S_l$ and H

Update v_a based on $N_a + 2(Amp - 0.5)S_a$ and H

end if

end for

end if

The experimental scenario is that in a room where exist a robot and a human, and the robot has three choices of action: to play with the human, explore alone or idle to take a rest. When the robot approaches the human, the human can reject the robot or accept it and engage in playful interaction with the robot. As shown in Fig. 5, the corresponding MDP model abstracts this situation to three places that the robot can choose to locate itself: a position near the human, a playground that the robot likes to explore, and a home position for resting. The robot can conduct three types of actions, which are IDLE, Go for human and EXPLORE alone. The states in the MDP describe the interaction states in an abstract way, which benefits the calculation and update of emotional dynamics. As shown in Fig. 5, the IDLE state represents the state of the robot staying at the home position and resting. Whereas, the Positive and Negative states reflect the types of influences that the resulting states of actions on robot emotions, and Positive represents successful results, while Negative can be seen as a failure. For instance, the Positive state means successfully playing with humans or successfully exploring alone, and the Negative state means

being refused by the human or cannot find interesting things while exploring. In the MDP, the rewards, also mentioned as basic weights in this project, that lead to the Positive state is positive, and negative for the Negative state. Both Go for human action and EXPLORE alone action can lead to these two states, depending on the results of actions. But only the IDLE action can lead to the IDLE state.

In the experiment, we compare two models: one is our designed decision-making model with emotion dynamics, and the other is based on our decision-making framework, but without the influences of emotional factors. In order to enable a direct comparison between the two models, we posit that in the default situation, the robot has the same scale values of WANTING, i.e. the sensitivity level of DOPAMINE to Satiation, towards states of playing with the human and playing alone at playground. The sensitivity of changes of DOMINANCE to stimuli, as well as the rewards for achieving these two states are also equal. For the scenario without intervening emotions, there are no “wanting” parameters, so we only set the rewards of the two aforementioned states to the same values. The two experiments are based on the

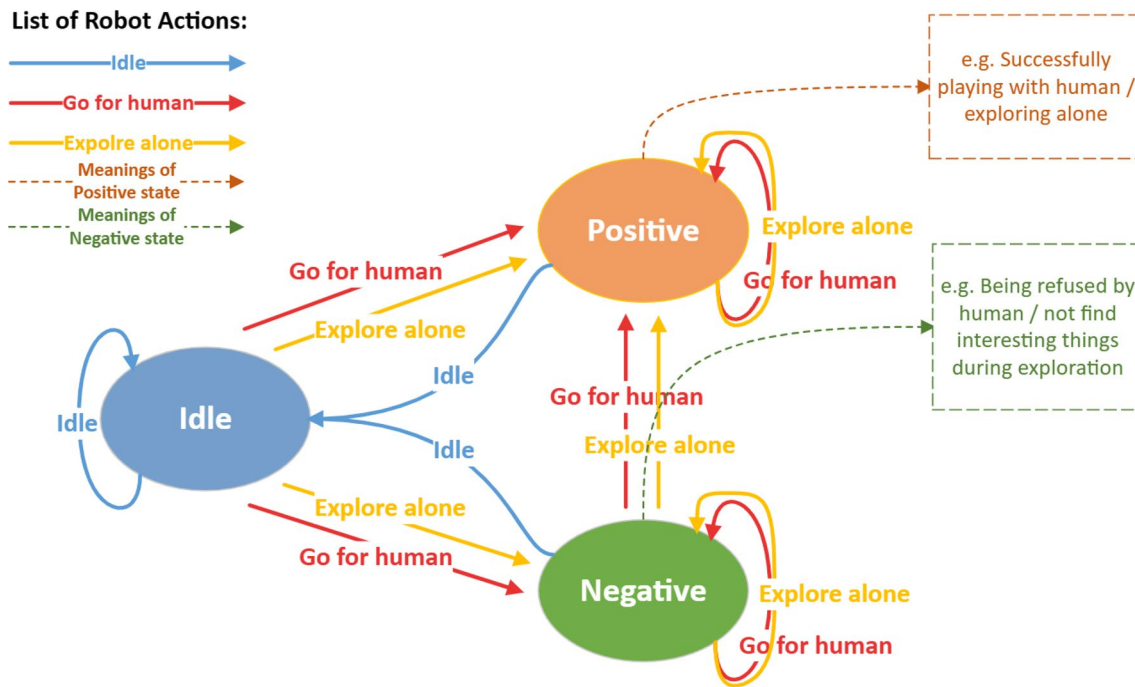


Fig. 5 MDP structure in the experiment. The experimental setup includes a room with a robot and a human. The robot has three choices of actions: playing with the human, exploring on its own alone, and idling or taking a rest. The states in the MDP describe the interaction states in an abstract way, measuring the influence that actions can

bring to emotional dynamics. For instance, the positive state represents a positive effect on robot emotion fluctuation with successes from playing with humans or exploring alone. Similarly, the negative state brings negative influence to emotions, and state Idle almost brings no changes to emotions

framework of decision making we proposed. Therefore, for the scenario without emotions, the results of interacting will influence the belief of the robot on the probability of success. While for the scenario with emotions, the interaction results will not only influence the belief of the robot regarding the probability of success from an aspect of interact history in reality, the accordingly variant emotions will also have influences on the probability as well as the weights (similar to rewards), illustrating the changes of the understanding to the world by the robot, and to what degree those states (e.g., being able to play with the human) are important to it.

4.1.2 Implementation

To evaluate the model performances in a changing environment, we consider the following experimental situation:

The human participant at first shows much interest in the robot, and continues interacting and playing with the robot for a while. But after that, the human refuses to play with the robot, and shows no interest in interacting with the robot.

The experiments of the decision-making module are conducted in simulation. The decision-making module will make decisions 60 times during each episode, and for each experiment group, we run 50 episodes to achieve average emotional values and probabilities of action conducted. In

each episode, the human accepts the robot and makes the robot play successfully with the human before the 15th state, as long as the robot conducts the Go for human action. After the 15th state, the human always refuses the robot if it turns to human. Under such situation settings, the experiment groups are divided according to combinations of emotional dynamics, as well as distinctive sensitivity of DOPAMINE and DOMINANCE, illustrated as follows:

- *Group 1:* Decision-making model without influences of emotional factors.
- *Group 2:* Decision-making model with only DOMINANCE and DOPAMINE, with equal sensitivities of them to states.
- *Group 3:* Decision-making model with only AROUSAL and DOPAMINE, with equal sensitivities.
- *Group 4:* Decision-making model with only DOMINANCE and AROUSAL, with equal sensitivities.
- *Group 5:* Decision-making model with all emotional factors in the model, namely, AROUSAL, DOMINANCE, and DOPAMINE, with equal sensitivities of DOMINANCE and DOPAMINE to states.
- *Group 6:* Decision-making model with all the emotional factors, and the sensitivity of DOPAMINE is higher towards human and lower towards exploration. This can be understood as easily having a stronger “wanting” to play with human, instead of exploring alone.

- *Group 7*: Decision-making model with all the emotional factors, and the sensitivity of DOPAMINE is lower towards human and higher towards exploration.
- *Group 8*: Decision-making model with all the emotional factors, and the sensitivity of DOMINANCE is higher towards human and lower towards exploration. This can be understood as the DOMINANCE to successfully play with the human is more easily to be influenced by what has happened, while the DOMINANCE to exploring is harder to influence.
- *Group 9*: Decision-making model with all the emotional factors, and the sensitivity of DOMINANCE is lower towards human and higher towards exploration.

Apart from the emotional factor A (AROUSAL), D (DOMINANCE), DA (DOPAMINE), the emotion P (PLEASURE) has no impact on robot decision making but is also calculated and updated according to changes of stimuli. As a result, it is also illustrated in the diagrams functioning as a reference to the mood of the robot.

For the decision-making model with emotional factors, we assess the realistic effects of emotions when the robot faces a change of environment. Theoretically, the function of AROUSAL is to reduce rational judgement according to interaction history, so the behaviours of robots will not converge very fast if the level of AROUSAL (A) is large. Only AROUSAL influences robots' understanding of success probability. The function of DOMINANCE is to increase their understanding weights to staying idle (rest/home) and reduce weights to move, no matter towards human or explore alone, if the level of DOMINANCE (D) is low. D has three types, the total D , D_1 (D for playing with human), and D_2 (D for exploring alone). They follow the same generate method, with different types of stimulus values taken into calculation. For example, D_1 only responds to stimuli related to humans, like being accepted by the human or being refused. The function of DOPAMINE is to regulate robots' behaviours, and prevent robots from repetitive actions so that it shows a more natural and vivid behaviour. The level of DOPAMINE (DA) influences reaction of the robot to rewards. When DA is large and the robot has a stronger "wanting", the weights will increase and vice versa. DA has two types: DA_1 (for playing with human) and DA_2 (for exploring alone). The changes of DA depend on a type of variables named Satiation. Satiation is set to -1 if the robot failed to get positive rewards from the actions because the robot did not seek for it. If the robot sought for the rewards but failed, the Satiation is set to 0 . Once the robot gets the positive rewards, (of course after seeking for it), the Satiation will $+2$. Thus, a loop of robot behaviours is built based on Satiation, to get rewards for one time and then take rest for roughly 2 time steps unless influenced by any other factors.

4.1.3 Results

The results of decision-making without emotional factors are shown in Fig. 6. Though not modified by emotions, the model used in Group 1 also utilises Thompson Sampling (TS) to regulate decisions and optimise obtained rewards. From the graph, it is clear to see the trend that the robot reduces the probability of taking action Go for human after the 15th state, when the human stops to interact with it, and increases actions of EXPLORE alone and IDLE.

Referring to the emotion model described above, we compare the emotionless model (Group 1) with the model with full emotions (Group 5) first. As shown in Fig. 7, the trends of action probability in Group 5 are more flat than Group 1, reflecting a feature of stability of the emotional decision-making model. Yet, the emotionless model may seem more rational and can optimise obtained rewards quickly, the affective model provides a pattern where the robot always tries to engage in interactions, without giving up once the environment changes negatively. To evaluate the impacts of single emotional factors, we compare Group 2 to Group 4 to fully applied emotional model (Group 5). As shown in Fig. 6, scales of action probability at model without AROUSAL is broader than the model with AROUSAL, and in Group 2 the robot learns and practises to get higher set reward more rapidly than Group 5, no matter at the short period after 0th state or after 15th state. It can be understood that Group 5 is more irrational than Group 2, which makes impulsive decisions because of AROUSAL. For the model without DOMINANCE, Group 3 has a higher peak value of action Go for human and lower peak value for action IDLE compared with the fully applied emotional model. In the meantime, the scales of Group 3 are also broader than Group 5. This illustrates that with DOMINANCE, the robot will be more conservative to try uncertain choices, and spend more time to rest, if the DOMINANCE level decreases and loses confidence. For the model without DOPAMINE, we can see that in Group 4, the front section of the figure of action probability is almost the same as the emotionless model, which converges quickly to the optimal choices before the 15th state. But after that, the probability of action Go for human goes through a sudden and quick drop, with a significant increase of the probability of action IDLE occurring at the same time. Group 4 possesses a "sentimental" feature while making decisions, which reacts strongly when facing unexpected failures, significantly reducing choices related to failure and staying idle. There is also a stage of decrease of idle from state 16 to 24 approximately, which can be seen as the expression of frustration and ends with the promotion of DOMINANCE. Thus, the function of DOPAMINE can be described here as regulating the decisions of the robot, and preventing endless seeking of one type of reward.

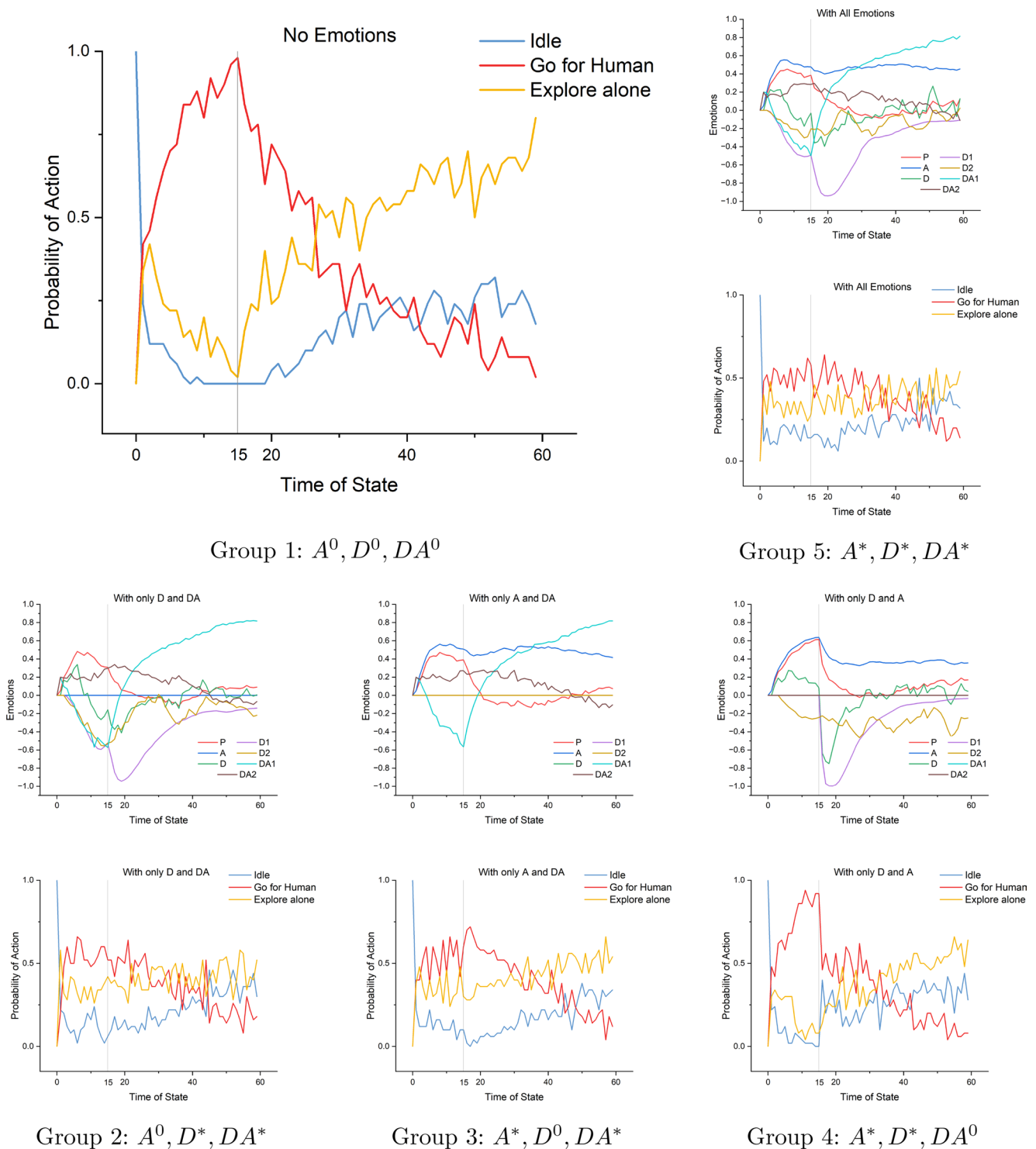


Fig. 6 Average emotional values and probability of taking actions for models only apply two out of three functional emotional factors. Superscript* represents validation of specific emotions and ⁰ means deficiency of emotions. Emotion PLEASURE does not engage in the decision-making and is applied here only for presentation, hence it exists in all the groups in this figure. The upper line of pictures illustrate that before the 16th state, the level of P increases and then decays a little bit, and afterwards with the human's always refusing, it decreases sharply and slowly rebounds a little after the robot becomes accustomed to the situation. The level of A and D follows a similar trend with P , but on different levels. DA_1 first generally drops and

then rises after the 15th state, reflecting boredom of playing with human at the initial stage and "wanting" to do this after the human refuses to play. The level of DA_2 fluctuates relatively stably, since the probability of getting rewards from EXPLORE alone never changes. However, there is usually a valley occurring around the 15th state. It should be noted that the changes of emotions are influenced by the selection of different emotion collections. For example, in Group 4 the vacancy of DA results lack of tiredness and "wanting" in the robot's behaviours, and the very "rational" actions prevents P and A decaying lately before the 15th state

Aside from testing models with only two functional emotional factors, we also explore the various influences of sensitivities of DOPAMINE and DOMINANCE towards Satiation and Stimuli in Group 6 to Group 9. As illustrated in Fig. 7, we apply higher sensitivity of DOPAMINE towards playing with human, and less sensitivity towards exploring in Group 6, with reversed ones in Group 7. Comparing the action probability of Groups 6 and 7, the peaks of action Go for human and IDLE in Group 6 occur later than Group 7, happening after the 15th state, while the peaks of Group 7 occur almost around the 15th state. This reflects a short-term insistence of going for human after being refused, when the DOPAMINE, or the “wanting” towards human is highly sensitive. The holistic probability of EXPLORE alone in Group 7 is also slightly higher than that of Group 6, reflecting the increase of taking such action under higher sensitivity of DOPAMINE towards it. The values of DOPAMINE towards human (DA_1) and DOPAMINE towards exploration (DA_2) are also shown in the upper line of the pictures in Fig. 7, illustrating an obvious increase of DA_1 in Group 6, caused by the combination of unsatisfied requirements and the highly sensitive feature of DA_1 .

In Fig. 8, the models with different sensitivity of DOMINANCE towards human and exploration are compared. In Group 8, the sensitivity of DOMINANCE to Stimuli towards human (D_1) is higher, and the one towards exploration (D_2) is lower, which are reversed in Group 9. The probability of action Go for human in Group 8 achieves a minor peak at around state 3, and after the 15th state, it levels off for about 10 states. While in Group 9, the probability of action Exploring alone achieves the minor peak at state 3, and the probability of action Go for human reaches a peak shortly after the 15th state, different from Group 8. The higher sensitivity of DOMINANCE makes the robot confident to choose choices when there is not much historical information that can be referred to, which leads to the minor peak in Groups 8 and 9 on action Go for human and EXPLORE alone. Whereas, when encountering failures, the models with higher DOMINANCE sensitivity are more easily to be frustrated, causing the levelling-off stage of Go for humans in Group 8. In addition to the emotions and probabilities of actions, we also evaluate the emotional influence on probabilities and rewards and demonstrate in a more intuitive way. Figure. 9 illustrates how AROUSAL influences the transition probabilities

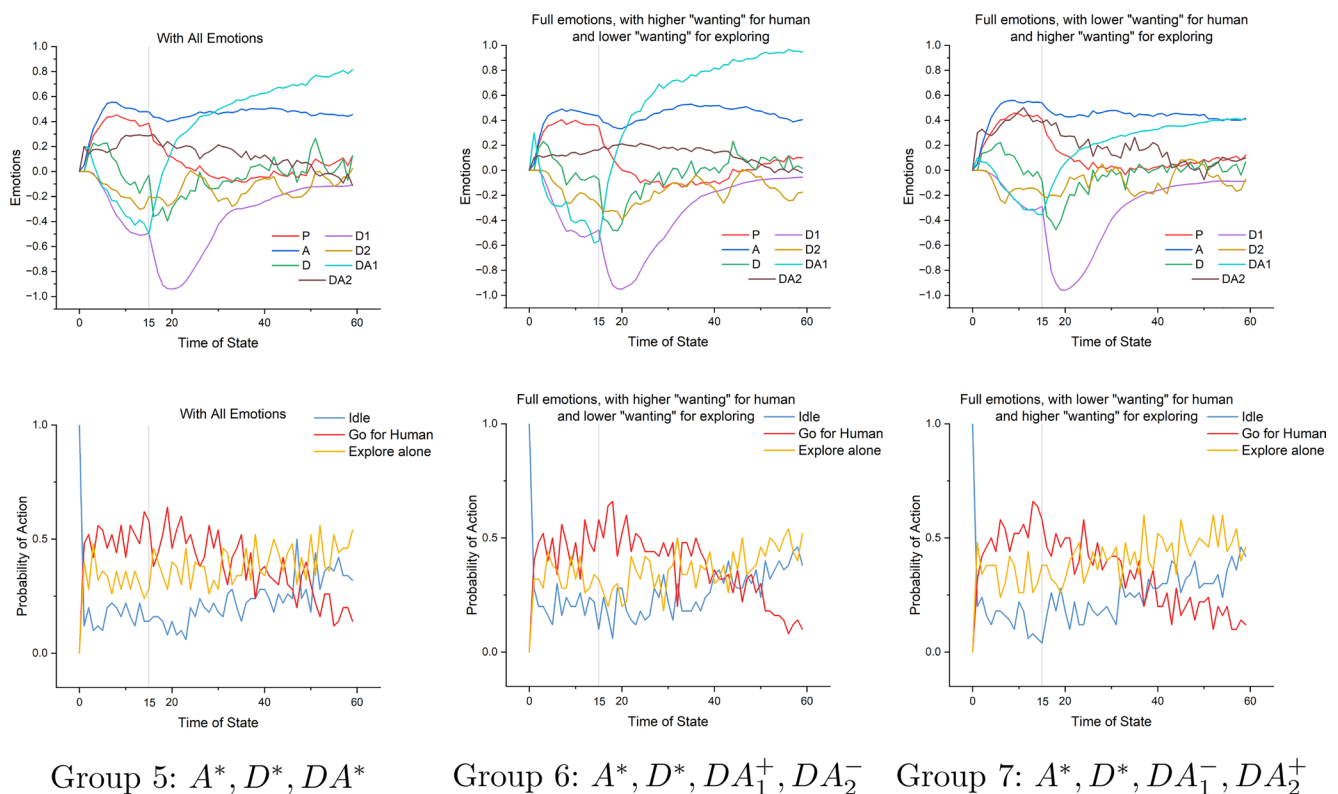


Fig. 7 Average emotional values and probability of taking actions for models with full emotions and different sensitivity of DA (Dopamine). Superscript “*” represents validation of specific emotions, with “+” and “-” meaning increase of sensibility and decrease of sensibility respectively.

As illustrated in the upper line of pictures, the level of DA with higher sensibility fluctuates with greater amplitude and vice versa. For instance, DA_1^+ in Group 6 has a deeper valley and ends up higher compared to DA_1^- in Group 7

in MDP. Comparing Group 2 with Group 5, with the joining of AROUSAL, all transition probabilities are adjusted to be closer to 0.5. Figure 10 shows the relationship between DOPAMINE, DOMINANCE and the rewards. By comparing Group 4 and 5, it's obvious that DOPAMINE is fundamental for reward regulation; Since in the experiments the standard reward sensitivity to DOMINANCE is set small, and Group 3 has not much difference with Group 5, we increase the sensitivity to show the function of DOMINANCE on rewards better, as shown in the last picture of Fig. 10. It's clear that during the time the level of DOMINANCE is significantly low, i.e. shortly after the 15th state, which causes the reward of taking action IDLE increase and the rest of rewards closer to 0, especially the rewards towards the Positive state.

4.2 Behaviour-Generating Module

4.2.1 Experimental Design

In the experiment of the Behaviour-Generating module, we evaluate the interpretability of robot-generated affective behaviours for humans by asking participants to watch videos of robot behaviours under different emotions. The scenario is demonstrated based on simulation, and the robot we use is MiRo, a bunny-like robot [51], which can move its head, neck, ears and tail, although not in a complex way. Based on the principle rules of the behaviour-generating module in Section 3.4, we simulate the scenario using ROS and Gazebo. In the experiment, the robots are programmed

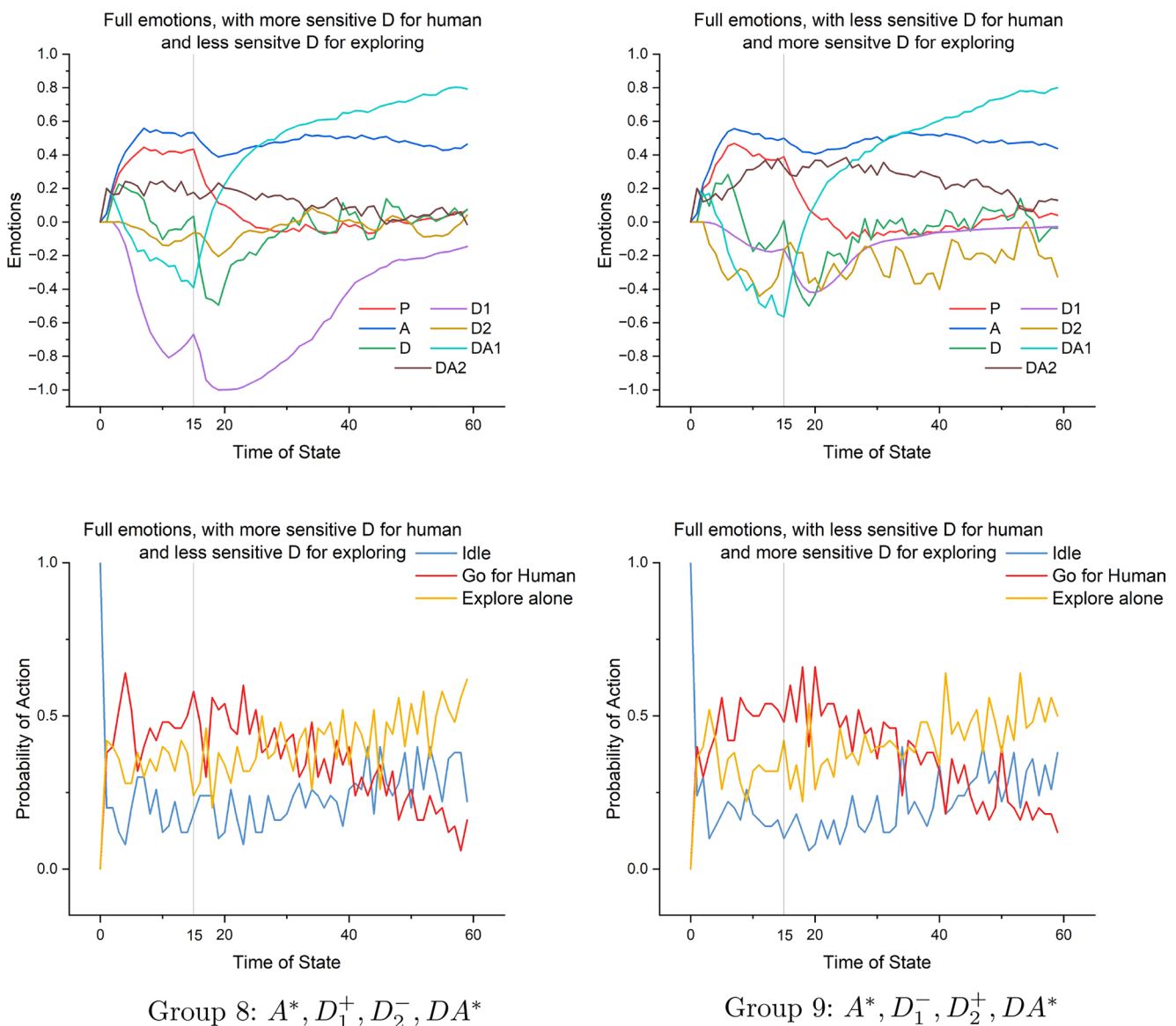


Fig. 8 Average emotional values and probability of taking actions for models with full emotions and different sensitivity of D (Dominance). Super-script * represents validation of specific emotions, with $+$ and $-$ meaning increase of sensibility and decrease of sensibility respectively

to move in a small piece of ground, and the standard behaviour for neutral emotions is circling slowly. Based on circling, the fundamental behaviour, the behaviours of the robot will vary and extend from circling according to their emotional dynamics. Since the functional emotional parameters in the behaviour-generating module are restricted to PLEASURE (P), AROUSAL (A) and DOMINANCE (D), the experiment groups are divided according to distinctive values of these three dimensions of emotion. There are three possible states for each dimension of emotional state in this experiment, which are positive state (emotional parameter value $\chi = 0.6$), neutral state (emotional parameter value $= 0$) and negative state (emotional parameter value $= -\chi$). For reference, the parameters P , A , and D have all the range $[-1, 1]$, with 1 standing for very pleased, aroused or confident, and -1 for very upset, sleepy, and lack of confidence. By combining these states, the experiment groups share the emotional values in 15 different combinations,

$$(P, A, D) = \begin{cases} (0, 0, 0), (\chi, \chi, \chi), \\ (-\chi, -\chi, -\chi), (\chi, \chi, 0), \\ (0, -\chi, 0), (0, \chi, \chi), \\ (-\chi, -\chi, 0), (0, \chi, 0), \\ (0, -\chi, -\chi), (\chi, 0, 0), \\ (-\chi, 0, -\chi), (0, 0, \chi), \\ (-\chi, 0, 0), (\chi, 0, \chi), \\ (0, 0, -\chi) \end{cases}$$

so that we have 15 groups in total (We set the parameter here always to $\chi = 0.6$). A set of examples of static affective behaviours with such kind of emotional value sets are shown in Fig. 11.

During the experiment, the human participants were required to watch 15 videos based on the aforementioned emotional sets. The sequences of the videos are randomised, and the participants can watch the videos multiple times if

needed. After watching each video, the participants will be asked to rate their impressions on the robot behaviours shown in the video in a questionnaire, from the aspects of PLEASURE, AROUSAL and DOMINANCE. In the design of the questionnaire, we apply the questionnaire Table 5, where the description of the three emotional states refers to the study [52] which measures human emotions on the aspect of P , A , D .

4.2.2 Implementation

We recruited 21 people (8 Male, 11 Female, 1 Other, 1 Unwilling to say) speaking English to participate in the experiment. Their knowledge and degree of familiarity with robots are that 38.10% of them know robots very well; 33.33% have some knowledge of robots; 23.81% know a little about robots; 4.76% barely know nothing about robots. The experiment is conducted in the form of an online survey, where participants first finish basic information, and then watch the 15 videos in a randomised sequence. During or after watching each video, the participants rate their impressions according to Table 5 immediately, then proceed to the next video.

4.2.3 Results

The results of each experiment group are processed and compared with the reference emotion values, i.e. the emotional values used to generate behaviours. The results of impression rating are also processed, utilising the mean values of each group to calculate the correlation. The Pearson correlation coefficients and p -values of human emotional impressions in the experiments and reference emotion values are illustrated in Table 6. We can see that the emotional dimension of PLEASURE matches the reference values best, with a Pearson correlation coefficient of 0.93, followed by

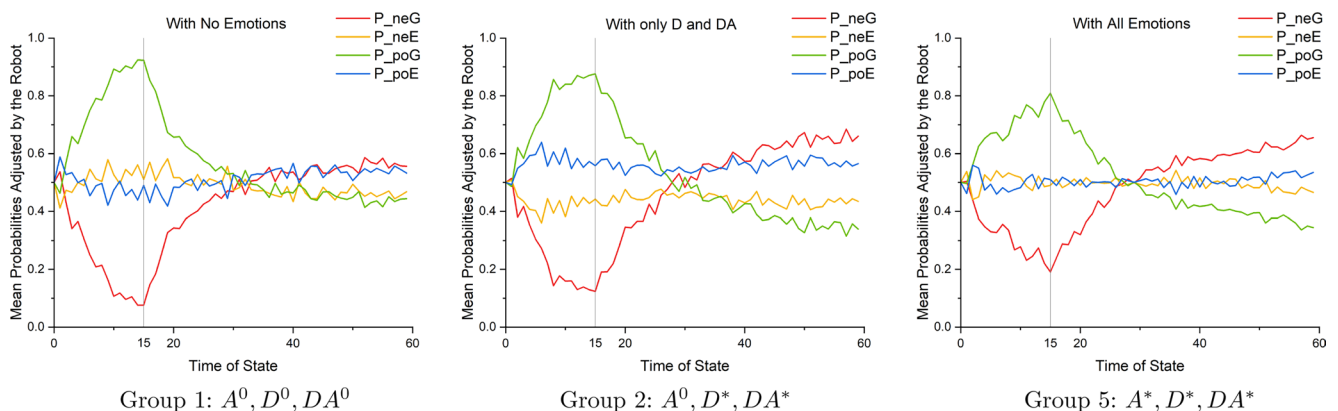


Fig. 9 Average transition probability adjusted by the robot in the MDP. “P” represents the transition probability, followed by the name of the state transitioned to and the name of actions that are taken. “ne” represents the state negative, and “po” represents the state Positive; “G” represents the action Go for human, and “E” represents the action explore

alone. For example, P_neG represents the transition probability from any state to state negative by taking action Go for human. The rest of the transition probabilities are either 0 or always stable, so only these four significant probabilities are shown in the pictures

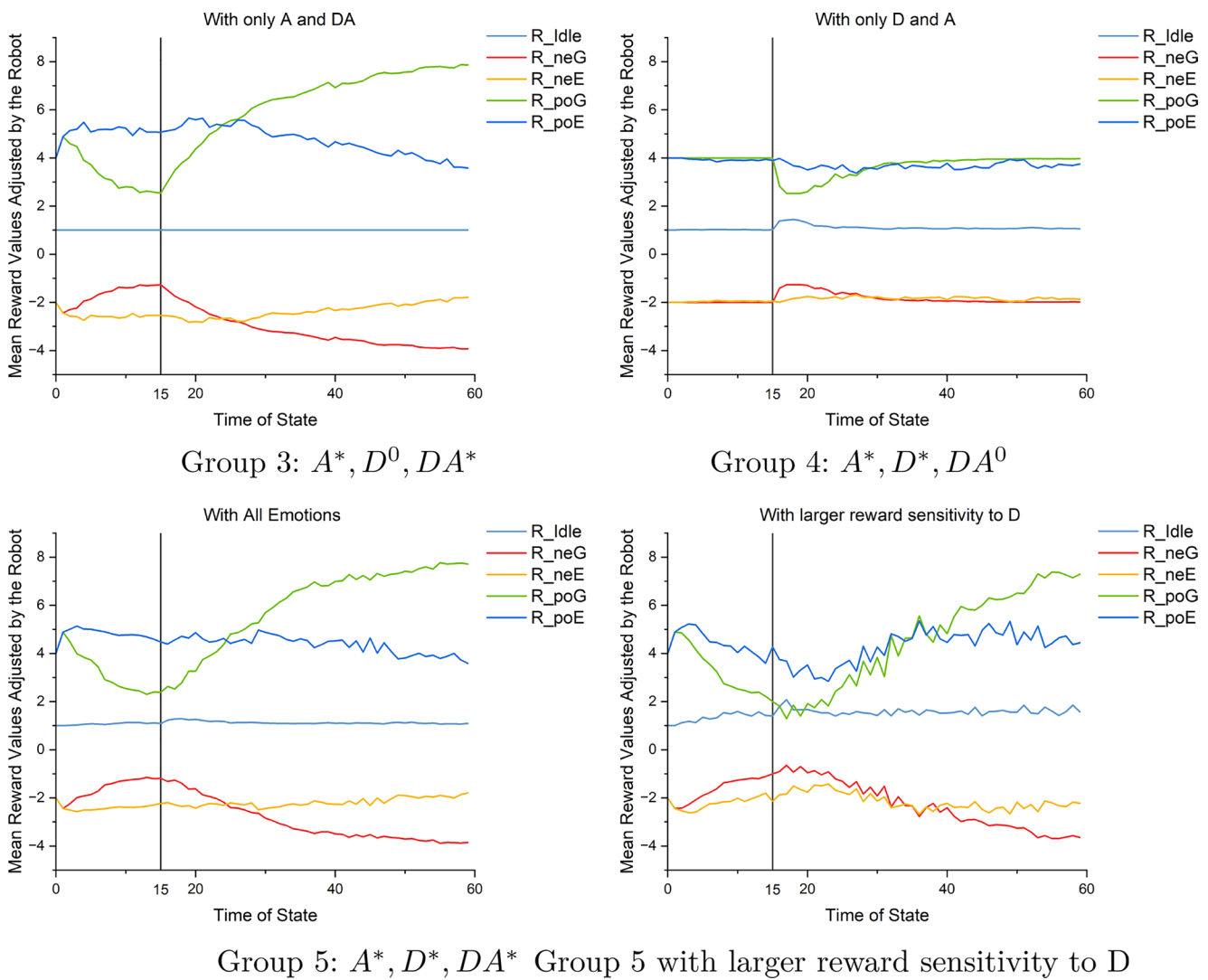


Fig. 10 Average reward values adjusted by the robot in the MDP. Similar to Fig. 9, “R” represents the rewards, followed by the name of the state transitioned to and the name of actions that are taken, except for R_Idle, representing the reward from any state to state Idle by taking action Idle

AROUSAL and DOMINANCE, with Pearson correlation coefficients of 0.69 and 0.58 respectively. The p -values also share the same trend, with PLEASURE having the smallest p -value < 0.0001 , followed by AROUSAL and DOMINANCE. The p -values of which are 0.004 and 0.022 respectively. The results show that human participants are very good at distinguishing the behaviours of the robot in regard to the PLEASURE dimension, and the results are highly accurate, using values in the generating system as references. The second-best emotional dimension that humans can recognise is AROUSAL, and the worst one is DOMINANCE. More information on the impression rating results is illustrated in Fig. 12.

In Fig. 12, the mean values of human impressions of each experiment group are illustrated as black dots. The 95% confidence band is also shown in the pictures. We can see that the data distribution of PLEASURE is balanced and fit linearly

well, meaning that people distinguish PLEASURE accurately. However, people usually overestimate the AROUSAL value of robot behaviours, and there exists an upward shift of the data distribution in the second graph. In the third graph, the distribution of DOMINANCE is narrow on the direction of y -axis, meaning that people are relatively not well at distinguishing behaviours on the aspect of DOMINANCE, compared with that of PLEASURE and AROUSAL.

There may also exist mutual influences between emotional dimensions. For example, when people observe a robot behaviour with both high values of PLEASURE and AROUSAL, they may tend to overestimate the emotional value on the aspect of DOMINANCE. In the first graph of Fig. 12, we observe the data dot which is obviously downward deviating from the band. It is the experiment group with reference values of $(P, A, D) = (0, -\chi, -\chi)$. In this group, people underestimate the value of PLEASURE, possibly being

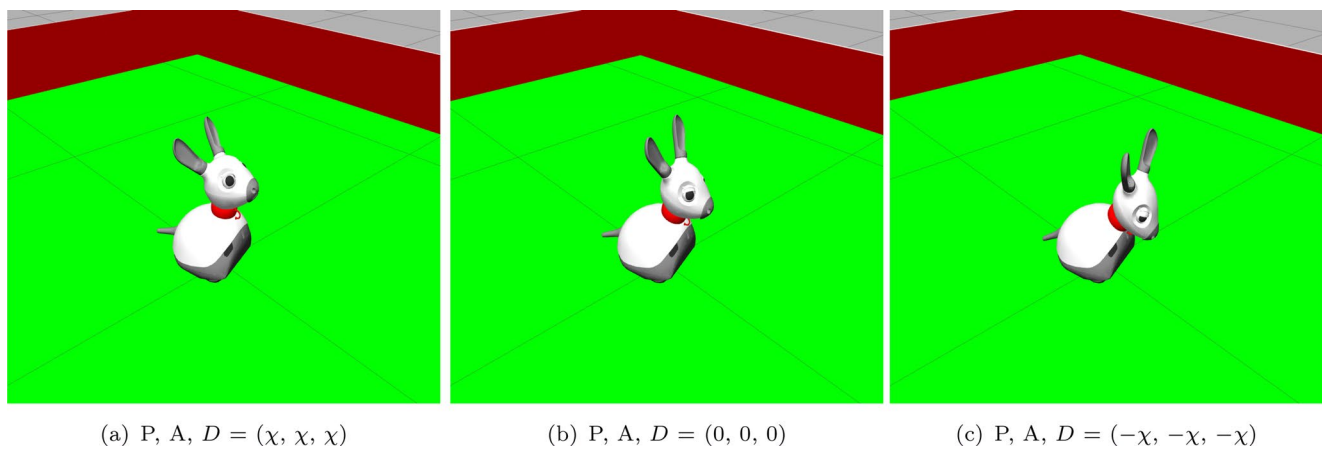


Fig. 11 Affective behaviour in the MiRo robot

Table 5 Questionnaire

Please rate your impressions of the robot behaviours on these scales:

Unhappy	1	2	3	4	5	Happy
Sleepy	1	2	3	4	5	Wide-awake
Controlled	1	2	3	4	5	Controlling

Table 6 Pearson correlation coefficients and p -values of human emotional impressions in the experiments

Emotions	Pearson Corr.	p -value
P	0.93	<0.0001***
A	0.69	0.004**
D	0.58	0.022*

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Bold value indicates statistically significant results

influenced by the joint low levels of AROUSAL and DOMINANCE. However, the PLEASURE value does not deviate in the group with values of $(P, A, D) = (0, \chi, \chi)$, so we assume that this influences of PLEASURE only happens when both values of AROUSAL and DOMINANCE are low. In the second graph, there is also a significantly deviated dot upward from the band, which is the group of $(P, A, D) = (\chi, 0, \chi)$. Similarly, the AROUSAL value does not deviate much in the group of $(P, A, D) = (-\chi, 0, -\chi)$, so we deduce that the PLEASURE and DOMINANCE have a strong joint influence on AROUSAL only when the values of P and D are large. Whereas, in the third graph, the pattern of influences to DOMINANCE is different. It seems that on the positive level, a large PLEASURE value itself can strongly influence the DOMINANCE value, given a neutral AROUSAL value. Moreover, a large AROUSAL value with the neutral value of PLEASURE also has a slight influence of DOMINANCE. The two dots that most significantly deviate from the band upwards and downwards are from groups of $(P, A, D) = (\chi, 0, 0)$ and $(0, 0, \chi)$. With the neutral AROUSAL, the single PLEASURE impacts people overestimate D value if high, and underestimate D value if low.

The groups of $(P, A, D) = (0, \chi, 0)$ and $(0, 0, \chi)$ also show a similar impact from AROUSAL to DOMINANCE in the picture.

5 Discussion

5.1 Affective Robotic Mind

Much work has been conducted on emotion recognition by robots as a pattern-matching problem or the expression of emotions by robots in human-robot interaction, see [53], whereas the integration of such abilities in the behavioural context and its interpretability has found less attention. For integrated systems based on feedback loops, the cited paper refers only to the work of Dietrich Dörner, known as Ψ -theory [54], and also to [55] who agree that “the ways in which humans and robots work together as a team” deserve more interest. The present study is more an attempt to show the complexity of the problem than a solution, but it also emphasises the possibility of separating the conceptual problem from the variety of useful developments in sensing, perception, planning, organisation of behaviour, control, learning, and evaluation. This is because the central organisation of a companion robot appears to be less easily realisable by utility-based machine learning approach [56], while a system-based affective architecture is not only natural, but tends also to be preferable by human users.

From the results of the decision-making module, we’ve already been able to see how complex decisions can be by just adjusting several parameters in the emotion model. Personalities of social robots may have a more simple solution

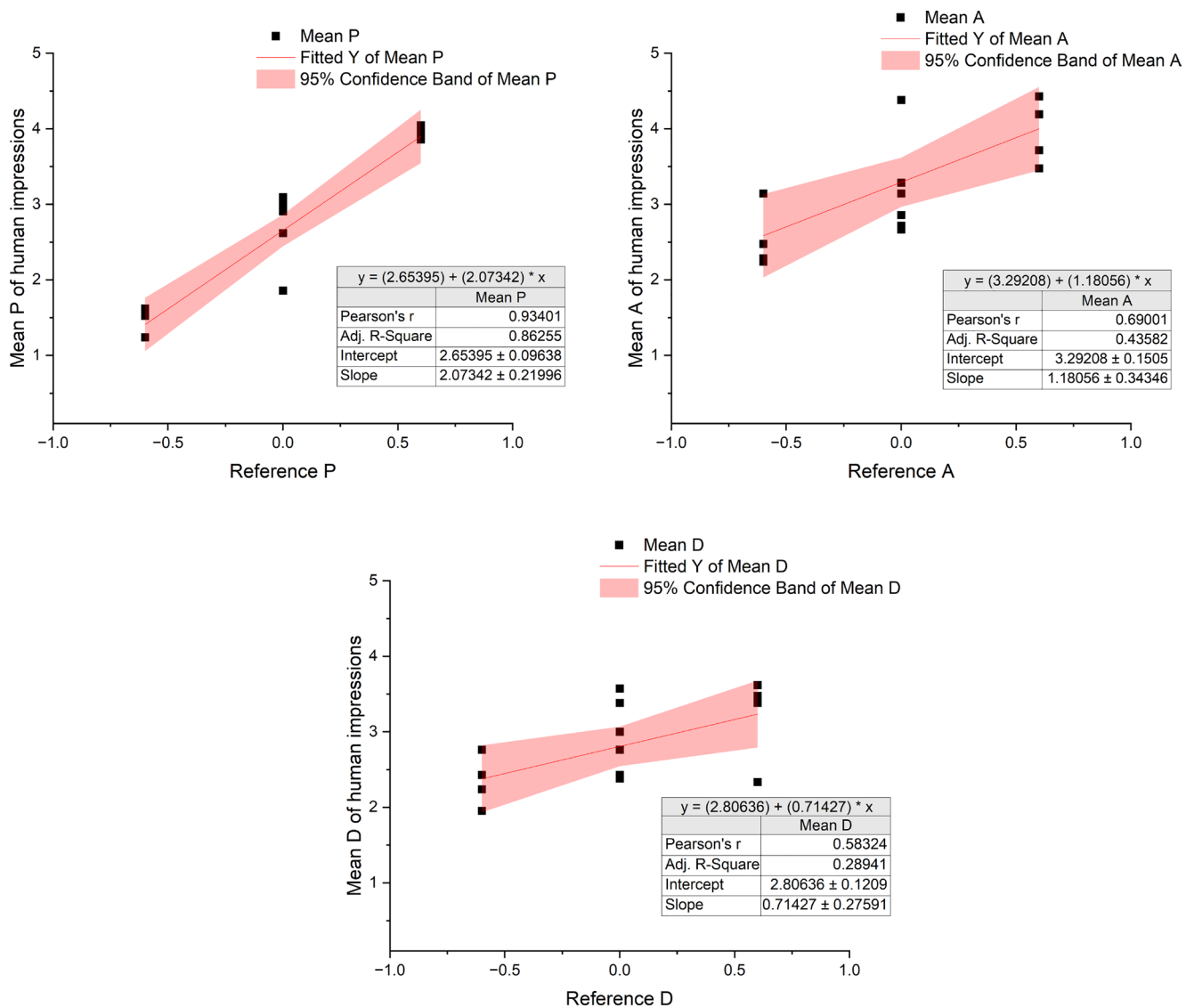


Fig. 12 Mean emotional impressions values of robot behaviours and the reference emotional values. For all three figures, the y -axis is for the mean values of emotions that humans perceive in the experiments, and x -axis is for the reference values of emotions

existing in this study. Although attributing variant values to the sensitivity parameters that control the influence of stimuli and satiety on emotional states, decision patterns like impulsive, conservative, sentimental and special preference for each choice are generated. This may shed light on robot personality design which usually aims at human satisfaction, and provides multiple possibilities to design robots for people with different preferences. Comparing to the decision-making process without incorporation of emotional factors, though introducing emotions can lead to the reduction in the efficiency of learning and completing tasks with the highest assigned rewards (at least shown in this paper), it has the potential in keeping constant and continuous exploration in long-term interaction with changing environment, regardless of how the rewards of tasks are

assigned initially by the human designer. It's also promising to be applied in the social scenarios of human-robot interaction, which may help promote humans' understanding of the reasons and contexts behind robots' decisions if combined with affective generated behaviours. The behaviour-generating module does not refer to the behaviours of any pre-supposed antetype organisms (such as rabbits for the MiRo robot). Humans are still capable of perceiving the emotions behind robots in this study, and the accuracy for distinguishing PLEASURE dimension of robot emotions is interestingly high. In addition, the results reveal possible correlations of P, A, D dimensions in human emotion perception. This may be caused by the limited positions of natural emotions in the PAD emotion space, and humans tend to categorise emotions aside from that into those natural ones. Further

work can be conducted and research whether this is a widely existing feature in human-robot interaction.

In the AROM model, we achieve the consistency and integration of decision-making and behaviour generation modules from the theory basis. The robot's decisions and behaviours are all regulated by the real-time emotional dynamics from the same emotion model, which guarantees the identical logic behind each robot move. Only when the decisions and behaviours are driven by the same emotions can we experiment with more advanced cognitive functions in human-robot interaction. For instance, by observing the robot's decision to grab an apple and physical behaviour of reaching hands with high velocity driven by one psychological status: anxiety, humans may be able to apply their Theory of Mind (ToM) functions on robots, and have a better understanding of the robot's intentions. In addition, the link of the stimuli that arouse changes of emotions with the reward in Reinforcement Learning also supports building a closed-loop emotion-decision-behaviour system, making the logic behind robot behaviours more fluent and continuous. However, the impact of the consistency and integration of robot affective decision-making and behaviour generation still needs to be evaluated and developed. Further experiments of the integrated system of AROM should be conducted to assess the dimensions and degrees of its impact on enhancing performances of human-robot interaction, such as the capability of humans to recognise robots' intentions and beliefs.

In the design of the AROM system, we aimed at enhancing the applicability and enabling generalisation to various robotic platforms. Design towards specific robot types and decision-making methods is avoided, and the core concepts in the model are easy to generalise, such as the velocity, amplitude and frequency features in behaviour generation, as well as the weight and probability in decision making. For other types of robots like humanoid robots, it's easy to build a match between the dynamic parameters of the robot and the affective features of AROM; and for other decision-making approaches, as long as they are based on the rewards/weights and probabilities, AROM model can make a good fit. Despite the design principle of applicability and the ability to be generalised, additional experiments are required in order to test the performance of AROM model on other robotic platforms and applications. It's possible that the settings in AROM cannot meet the requirements for realisation on some platforms and scenarios.

5.2 Limitations

The MiRo robot used in this study does not have many expressive options and the variety of movement patterns for expressive behaviour is limited. This situation was suitable for the present study, in line with the MiRo robot being designed for specific use cases, but requires a considerable extension, if direct and unsupervised applications of pet-like robots in educational or therapeutic situations are to be realised and, in particular, if a development towards commercialisation is intended.

The experiments on behaviour generation may raise cognitive fatigue of human subjects, since they are required to watch 15 videos per session. Although each video only lasts for around 10 seconds and all of the videos takes less than two minutes to watch, the repeated presentation of videos may cause fatigue of the participant's attention. This can have a potential influence on the experimental results, but is mitigated by the randomisation of the sequence of the videos.

Because only subjective decision making is considered here, the evaluation does not include the realism of the robot behaviour, its compliance with expectations, and efficiency of the behaviour, which is a problem in particular in the presence of an unpredictable human. Although some of these aspects have been addressed in the experiments included here, the design of the experiments is biased toward simplicity and assessability which may not be indicative for the general richness of human interaction. Although only basic emotions and physiological factors were studied, it is still a valuable start that contributes to our understanding of the ways that human-robot interaction can develop. Further studies will focus on more complex emotion structures and more complex interactions among the modules of the AROM model.

We should also mention that the system is based on MDP whereas could appear to be more suitable for a realistic situation where many aspects of the mutual affective contexts are not directly observable which is explicitly covered in partially observable MDPs (POMDPs) [57]. Here we need to choose among several options, whether the limitations of the continued use of an MDP are acceptable, whether a probabilistic sensor model is sufficient, whether more information is to be acquired by a wider action set, or whether it is sufficient to work with a small number of hypotheses in parallel. Similar options have

been studied in the field of POMDP, but a decision is possible only in a practical setting.

Ethical questions related to the project have been addressed in the context of an Informatics Ethics Application which has been approved. However, long-term effects that are explicitly excluded from the present study, can be expected to be more critical. In the long-term interaction with a robot, well-meaning design features may have effects opposite to the original intentions, especially for an AI-based learning robot. For example, robots' support to humans could affect lower levels of activity in the human, or it could become an annoyance, if the robot has learned patterns from the human behaviour that would preferably be kept private. These aspects are part of the on-going debate about the limits of AI, but are beyond the scope of this study. It is, however, possible to predict that features of the current design, such as interpretability as a main characteristic of the robot mind will be useful also in the more general situation of use of such a robot in practice. Nevertheless, with the adoption of robots as companions, we are entering here uncharted territory that requires broader and deeper insights before it can be entered.

So far the experiments carried out in this paper are still limited, with further experiments and more systematic assessments required in order to evaluate AROM features of integration of decision-making and behaviour generation, as well as its ability to be generalised to robot platforms other than MiRo and other interaction scenarios. The advantages and limitations of these model features should be analysed, for the sake of promoting broader and more advanced applications.

5.3 Future Work

Further experiments can be implemented based on the present study. To explore the boundaries of AROM system, other types of robots such as humanoid robots can be applied. It is also beneficial to conduct experiments on the combination of the decision-making and behaviour-generating modules, evaluating the joint interpretability of decisions and behaviours under the same affective context. Research concentrated on the influences of DOPAMINE and "wanting" is also meaningful, which may fill the gap of human perception of

robot intentions from an aspect of emotion. Furthermore, since the nature of the AROM emotion model is to build connections between stimuli and emotions, advanced exploration of "stimulus-reinforcement association" [9] may enable more complex behaviours in robots. For instance, by learning the association between the stimulus PLAYGROUND with the emotional reinforcer PLAYING, affective robots can learn to get excited when they encounter stimuli related to PLAYGROUND. It is to be researched whether the emotion models under this architecture can help shape robot personalities and behaviour logic to enable advanced human-robot interaction.

6 Conclusion

Overall, the contribution of this paper was to provide an architecture for robots to conduct appropriate actions in the context of a human-robot companionship. The system allows for a concurrent and adaptive diversity of behaviours, including the tendency of the robot to explore its environment, execute actions for its own maintenance, and engage with a human partner. This biomimetic approach to behaviour generation may provide a reference for a generic robot where copying of seemingly similar animal behaviours is not an option or for the design of a specifically animal-like robot that needs to include interpretability and adaptability within to context of a potentially vulnerable human partner. The affective model proposed here could also be useful for robot-assisted support for autistic children, since it includes more dimensions such as WANTING compared to traditional models with core emotions only. Likewise, a framework is suitable by design as a companion robot for elderly people, as a social robot in various contexts or simply as a toy. It remains to be tested whether the proposed framework would be suitable also for a companion of solitary pet animals.

Values of Parameters in Experiments

The values of parameters in the numerical experiments are shown in Table 7.

Table 7 Values of parameters in the numerical experiments

Parameters	Values
Parameters for emotional dynamics and interactions	
R	See W_{bs} and $W_{bs-idle}$
S^A	+2 at positive feedback, -1 if not engage, 0 at negative feedback
k_{da}	$e^{-k_{da}} = 0.8$
k_p	$e^{-k_p} = 0.7$
k_a	$e^{-k_a} = 0.7$
k_d	$e^{-k_d} = 0.7$
m_p	0.05
m_a	0.05
m_d	0.15
κ	-0.2
h	0.5
q	6
χ	0.6
Parameters for decision-making	
W_{bs}	For both action Go for Human and Explore Alone, if the result is positive then $W_{bs} = 4.0$, and if negative then $W_{bs} = -2.0$
$W_{bs-idle}$	1.0
λ	1
μ	0.5
δ	0.5
U	0.5
P	The initialised P for both Go for human and Explore Alone actions to both positive and negative states is 0.5. The transfer probability for Idle action to Idle state is 1, and to other states is 0.
Parameters for behaviour generation	
H_{Sp}	1
H_{Sd}	1
H_{Va}	1
H_{Vd}	1
H_{Ap}	2
H_{Aa}	1
H_{Ad}	1
H_{Fa}	1
v_{Sd}	1
v_{Vd}	1
v_{Ad}	1
v_{Va}	0.5
v_{Aa}	0.5
v_{Fa}	1
K	When $A + P \geq 0.2$

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Data Availability <https://github.com/Jellyfishall/AROM>

Declarations

Ethics Approval This study was conducted according to relevant University of Edinburgh, national and international guidelines. The Informatics Ethics Application 583,562 covered all experiments in this study and was approved by the School of Informatics Ethics panel on 2. Aug. 2023.

Consent to Participate All the participants in this study have signed an informed consent form accepting to participate in the study.

Consent to Publication All the participants in this study have signed an informed consent form accepting to publish the anonymously obtained data.

Conflict of Interest All authors disclosed no conflict of interest.

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